Adaptive Heterogeneous Support Tensor Machine: An Extended STM for Object Recognition Using an Arbitrary Combination of Multisource Heterogeneous Remote Sensing Data

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***Abstract*— Multiple observation data from multisensor satellites lead to data with complex heterogeneous tensor structures, and the available object data are observed by the arbitrary combination of multisource satellites, which brings significant challenges to object recognition using typical deep learning methods, classical support vector machines (SVMs), and support tensor machines (STMs). To process multisource data represented as heterogeneous tensors effectively, the heterogeneous STM (HSTM) is proposed by integrating rank-*R* decomposition of multitensors with nu-STM to directly separate samples in heterogeneous tensor space. Furthermore, combining the shared projecting tensors used to mine related information for intercombinations and the auxiliary projecting tensors exploited to capture the individual information for intracombination, the HSTM is extended to Adaptive HSTM (AHSTM) to address the arbitrary combination of multisource data by only training once. To accelerate the training of AHSTM, the AHSTM-oriented decomposition method is proposed to use small sequential analytic optimizations instead of the original large numerical optimization. Using multiangle optical and synthetic aperture radar (SAR) satellite images, experimental results demonstrate that the proposed AHSTM obtains higher accuracy than typical SVM and STM methods for each combination of multisource data and reduces the time consumption by over 75% for the training set with different sizes compared with the typical interior-point method and the active-set method.**

***Index Terms*— Adaptive heterogeneous support tensor machine (HSTM), adaptive HSTM (AHSTM)-oriented decomposition algorithm, arbitrary combination of multisource data, multisource data collaboration, object recognition.**

I. Introduction

With the rapid development of satellite and sensor technology, multiple observation data embedded with complementary spectral, spatial, and context information acquired from multisensor satellites can be obtained to support more effective object recognition [1]–[3]. Since different objects can be observed by satellites with a single type or multiple types of sensors, object recognition technology must address multisource object data in addition to single-source object data.

Various recent studies on object recognition have mainly been based on deep learning networks. In [4], combined with feature pyramid networks and the dimension transformer module, local attention networks were constructed to detect occluded objects in optical remote sensing images (RSIs). By integrating backbone Tiny-Net and global attention block, remote sensing region-based convolutional neural networks are built to detect tiny objects in large scene optical RSIs [5]. Using the Faster-convolution neural network (CNN) [6] framework, a deep learning-based object recognition method is established to extract inshore and offshore ships in synthetic aperture radar (SAR) images [7]. By virtue of the complementary information of multisource data, the multisource region attention network-based classification method is developed to improve classification results using optical and light detection and ranging (LiDAR) remote sensing data [8]. Although the above deep learning-based methods worked well for different object recognition applications, they were extremely reliant on an extensive training set. At the same time, the high cost of acquiring multisource data makes it difficult to collect a sizeable multisource training set to support the training of the model. In comparison, classical machine learning-based classification algorithms have a sound theoretical foundation, resulting in relatively high recognition accuracies and excellent generalization capabilities for different object recognition tasks using a small number of samples. Among the machine learning-based classification algorithms, the most prominent representative was the support vector machine (SVM), which has been successfully applied in various pattern recognition fields owing to its theoretical and computational merits. After the original SVM was invented by Vapnik and Lerner [9], a series of SVM extensions were constructed in recent decades. Since the original SVM outputs no results if the data cannot be separated completely by a hyperplane, slack variables are embedded into *C*-SVM [10] (i.e., the standard SVM) to ensure that the solution can always be obtained by tolerating classification error. To improve the classification accuracy for class-imbalanced datasets, the weights of penalty parameters were introduced into the weighted SVM [11] to tradeoff misclassification costs for different classes. To directly obtain

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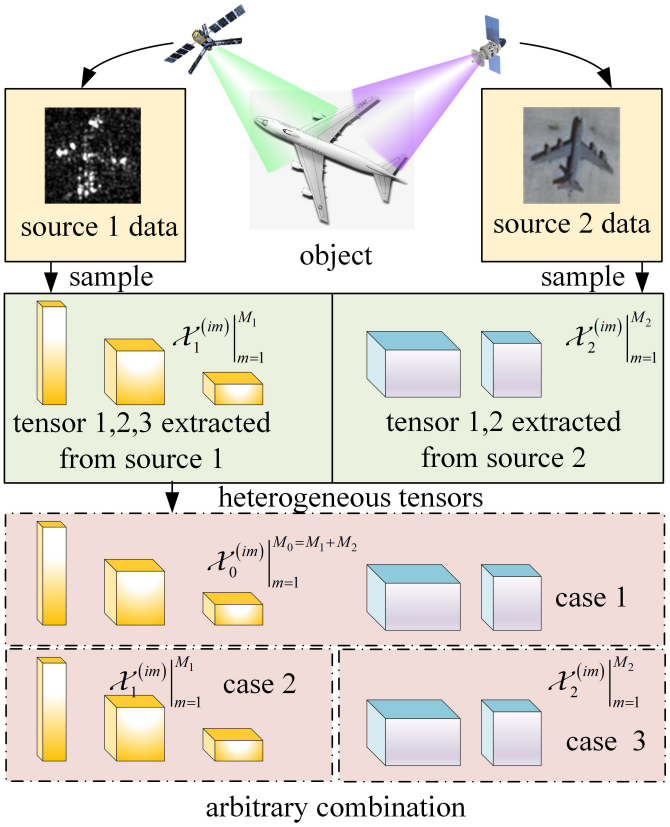


Fig. 1. Illustration of object recognition using multisource satellites data.

the analytic solution of the classification model, the least-squares SVM (LS-SVM) [12] was constructed to change the original inequality constraints of SVM to equality constraints to speed up the model training, while it damages the sparsity of the solution for standard SVM. To effectively control the classification error for the training set, ν-SVM [13] replaced parameter C of the C-SVM with parameter ν to limit the number of support vectors and errors. To reduce the impact of parameter C set manually on the classification results, the X-SVM was established to improve the robustness of the model to parameter C by introducing the separation efficiency criterion [14]. Although the above SVM and its variant algorithms have obtained effective classification results for specific tasks, they cannot directly process multisource remote sensing data naturally represented by tensors and have to convert them to vectors to comply with their input requirements. The existing research demonstrates that the conversion can lose useful information from the original data and increase classification errors [15]–[17].

To avoid vectorization conversion for input data, tensor-based classification algorithms are constructed to directly process samples in tensor space. Based on supervised tensor learning theory [15], the linear projection in vector space of SVM was replaced with the multilinear projection in tensor space to extend the C-SVM and ν-SVM to C-STM and ν-STM, respectively, [18]. In addition, the LS-SVM method was generalized to the LS-support tensor machine (STM) [19] to accept input data with a tensor structure. To avoid the nonconvex optimization of standard C-STM, the linear support higher order tensor machine (SHTM) was constructed by integrating the merits of C-SVM and tensor rank-one decomposition [20]. By utilizing the different kernel functions, the multikernel STM [21] (MK-STM) was established to improve the classification performance of linear STM. In addition, the multilinear rank STM (MRSTM) [22] exploits the Tucker decomposition [27]-based collection of projecting tensors instead of the original rank-1 projecting tensor to generalize the original STM. After applying multilinear principal component [24]-based preprocessing, the multiclass STM is developed to classify hyperspectral RSIs [23]. To attack the multiple objects classification tasks, the support multimode tensor machine (SMTM) leveraged the correlation between different classes by applying the multimode product in tensor space to output the classification results [25]. Combined with the rank-1 decomposition of the tensor, the low-rank approximation-based transductive STM (TSTM) was constructed to address the semisupervised classification problem [26].

Considering the practical situation in which the same object is observed by satellites with S types of sensors multiple times, the resulting object data will present various physical properties, which can be naturally represented as multiple tensors with different orders and dimensions. In this article, the set of tensors is defined as heterogeneous tensors, of which the number of tensors in the set is greater than one, and each tensor has the same or different orders and dimensions. Moreover, it is difficult to ensure that all of the objects can be observed by S-source satellites. In most cases, the obtained data can be considered to be an arbitrary combination of S-source data, as illustrated in Fig. 1. For convenience and without loss of generality, the number of types of sensors on satellites is set to 2 in this article. Therefore, there are three cases, of which cases 1, 2, and 3 denote the acquired data observed by satellites with two types of sensors, the acquired objects of data observed by satellites with sensors of type 1, and the acquired data observed by satellites with sensors of type 2, respectively. Although the STM generalized from SVM accepts the tensor as input, most of the existing STM methods can only process the input represented by a single tensor and cannot address the heterogeneous tensors directly, let alone process arbitrary combinations of multisource heterogeneous tensors. Facing this situation, the existing SVM and STM methods have to permute the multisource heterogeneous tensors to comply with their input requirement and train multiple classifiers separately to address each possible combination of multisource data, leading to damage of structural information for multisource data and the loss of related information between different combinations.

To achieve effective object recognition for arbitrary combinations of multisource data, two important improvements are essential for the existing STM methods, i.e., 1) extending the STM to accept heterogeneous tensors as input and 2) extending the STM to adapt to arbitrary combination situations. To process heterogeneous tensors, the approach taken is to

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include the utilization of multitensor rank-R projecting mapping instead of single-tensor rank-1 projecting mapping of ν-STM to directly separate samples in heterogeneous tensor space. To address arbitrary combinations of multisource data, different from the existing methods that train classifiers for each combination separately, the approach taken is to construct a single classifier and use samples from different combinations to jointly learn the shared classification hyperplane parameters to adapt to arbitrary combination situations. Following these two ideas, the heterogeneous STM (HSTM) is proposed to effectively recognize objects represented as heterogeneous tensors, and then, the adaptive HSTM (AHSTM) method is further proposed to extend HSTM to adapt to the arbitrary combination situations. In addition, to ensure the efficiency of training the AHSTM, the dual problem of the AHSTM-oriented decomposition algorithm is presented to break the complex numerical dual problem of AHSTM into a series of small analytic optimization problems. Finally, to demonstrate the superiority of the proposed methods, three datasets containing multiangle images and optical SAR images are utilized to evaluate the AHSTM and the corresponding decomposition algorithm in terms of classification accuracy and performance time, respectively.

The remainder of this article is organized as follows. Section II introduces some preliminaries, i.e., the basic tensor algebra and the classical SVM and STM. In Section III, the proposed HSTM, AHSTM, and the corresponding training methods are given sequentially. In Section IV, the decomposition method is proposed to accelerate the training of the HSTM and AHSTM. In Section V, the classification effect of the AHSTM is analyzed from the perspective of transfer learning (TL) and feasible regions. In Section VI, the parameter settings of HSTM are analyzed, the effects for HSTM and AHSTM are evaluated and compared with the typical SVM, STM, and deep learning methods, and the performance of the proposed decomposition-based dual problem of the AHSTM solving method is examined and compared with typical optimization solvers (i.e., the interior-point method and the active-set method). Our conclusions are given in Section VII.

II. Preliminaries

Before presenting our work, a brief introduction to some notations and fundamental concepts for tensor algebra and the classical ν-SVM and ν-STM methods is given in this section.

A. Tensor Algebra

Tensor algebra, also known as multilinear algebra, is an extension of linear algebra to multiway arrays. Following the convention for notions in tensor algebra [27], the scalar, vector, and matrix are denoted as lowercase letters (e.g., x and y), lowercase boldface letters (e.g., $\boldsymbol{x}$ and $\boldsymbol{y}$), and uppercase boldface letters (e.g., $\boldsymbol{M}$), throughout this article, respectively. Formally, an N-order tensor is denoted as $\mathcal{X} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$, where N represents the number of orders for the tensor and I_N indicates the dimension of the Nth order for the tensor. Some definitions of tensors related to the proposed method are displayed as follows.

Definition 1 (Outer Product): The outer product of M vectors (i.e., $\{\boldsymbol{a}_m \in \mathbb{R}^{I_m}\}_{m=1}^{M}$) is denoted as $\mathcal{A} = \boldsymbol{a}_1 \circ, \ldots, \circ \boldsymbol{a}_M$, whose results are the M-order tensor $\mathcal{A} \in \mathbb{R}^{I_1 \times \cdots \times I_M}$ with its $(i_1, \ldots, i_M)$ entry given by the following equation:

$$\mathcal{A}(i_1, \ldots, i_M) = \boldsymbol{a}_1(i_1) \times \cdots \times \boldsymbol{a}_M(i_M). \tag{1}$$

Definition 2 (Inner Product and Norm of Tensors): The inner product between two tensors with the same sizes (e.g., $\mathcal{A} \in \mathbb{R}^{I_1, \ldots, I_M}$, $\mathcal{B} \in \mathbb{R}^{I_1, \ldots, I_M}$) is defined as the sum of the products between their corresponding elements, that is,

$$\langle \mathcal{A}, \mathcal{B} \rangle = \sum_{i_1=1, \ldots, i_M=1}^{i_1=I_1, \ldots, i_M=I_M} \mathcal{A}(i_1, \ldots, i_M) \times \mathcal{B}(i_1, \ldots, i_M). \tag{2}$$

The Frobenius norm of the tensor is defined as follows:

$$\|\mathcal{A}\|_F = \sqrt{\sum_{i_1=1, \ldots, i_M=1}^{i_1=I_1, \ldots, i_M=I_M} \mathcal{A}(i_1, \ldots, i_M) \times \mathcal{A}(i_1, \ldots, i_M)}. \tag{3}$$

Definition 3 (Mode-k Product): The mode-k product of a tensor $\mathcal{A} \in \mathbb{R}^{I_1 \times \cdots \times I_M}$ with a matrix $B \in \mathbb{R}^{I_k \times I_k'}$ is denoted as $\mathcal{A} \times_k B$, whose results are a tensor of dimensions $I_1 \times \cdots \times I_k' \times \cdots \times I_M$, as expressed in the following equation:

$$(\mathcal{A} \times_k B)_{I_1, \ldots, I_{k-1}, q, I_{k+1}, \ldots, I_M} = \sum_{P=1}^{I_k} \mathcal{A}(I_1, \ldots, I_{k-1}, q, I_{k+1}, \ldots, I_M) \times B(p, q). \tag{4}$$

B. Support Vector Machine

As an effective variation of standard SVM, ν-SVM can conveniently control the number of support vectors and errors. Given the training set contained N samples $\{\boldsymbol{x}_i, y_i\}|_{i=1}^{N}$ where $\boldsymbol{x}_i \in \mathbb{R}^{I_1}$ denotes the feature vector of the ith sample, and $y_i \in \{1, -1\}$ indicates the corresponding label, the goal of the ν-SVM is to learn the classification hyperplane that separates the samples with different labels effectively, as shown in the following equation:

$$\begin{aligned} \min_{w,p,b} \ & \frac{1}{2}\boldsymbol{w}^T\boldsymbol{w} + \frac{1}{N}\sum_{i=1}^{N}\xi_i - \nu p \\ \text{s.t. } & y_i\left(\boldsymbol{x}_i{}^T\boldsymbol{w} + b\right) \geq p - \xi_i, \quad 1 \leq i \leq N \\ & p \geq 0 \\ & \xi_i \geq 0, \quad 1 \leq i \leq N \end{aligned} \tag{5}$$

where $\boldsymbol{w}$, b, p, ξ_i, and ν denote the normal vector to the classification hyperplane, the bias, the classification margin, the slack variable, and the parameter that is used to control the number of support vectors and errors, respectively. The research shows that the term $p \geq 0$ is redundant and can be removed during optimization [30]. To solve (5), the Lagrangian dual problem can be drawn as follows:

$$\min_{\alpha} \frac{1}{2}\boldsymbol{\alpha}^T Q \boldsymbol{\alpha}$$

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TABLE I
COMPARISON OF ACCEPTABLE INPUT DATA FOR DIFFERENT METHODS

input data	SVM	STM	HSTM	AHSTM
vector	√	√	√	√
single tensor	×	√	√	√
heterogeneous tensors	×	×	√	√
arbitrary combination of heterogeneous tensors	×	×	×	√

$$\text{s.t.} \sum_i y_i \alpha_i = 0$$
$$\sum_i \alpha_i = \nu$$
$$0 \le \alpha_i \le \frac{1}{N}, \quad 1 \le i \le N \tag{6}$$

where $Q(i, j) = y_i y_j \langle \boldsymbol{x}_i, \boldsymbol{x}_j \rangle$. After solving the above quadratic programming (QP), the optimal w and b can be calculated by the obtained $\boldsymbol{\alpha}$.Then, the label of test sample x can be predicted in the following decision function:

$$y(x) = \text{sign}\left(\boldsymbol{x}^T \boldsymbol{w} + b\right) \tag{7}$$

where sign$(\cdot)$ denotes the sign function.

C. Support Tensor Machine

The standard SVM accepts only vectors as input, whereas many data used in different applications are represented naturally as multiway arrays (i.e., tensors). Regarding object recognition using RSIs, both the raw RSIs and the extracted features can be considered to be tensors. For example, the optical image can be regarded as a third-order tensor, and the extracted mathematical morphology feature [43] can be described as a second-order tensor. When the data to be processed present a tensor structure, the SVM has to convert the data to a vector to comply with the requirement of input. To avoid the loss of structural information for original input caused by data conversion, the STM method is extended from SVM to classify the data in tensor space directly. Given training set $\{\mathcal{X}_i, y_i\}\Big|_{i=1}^{N}$ where $\mathcal{X}_i \in \mathbb{R}^{I_1 \times \cdots \times I_M}$ denotes the feature tensor of the ith sample, the parameters of ν-STM can be obtained by solving the following optimization problem:

$$\min_{\boldsymbol{w}_i \mid_{i=1}^{M}, p, b} \frac{1}{2} \|\boldsymbol{w}_1 \circ, \ldots, \circ \boldsymbol{w}_m\|_F^2 + \frac{1}{N} \sum_{i=1}^{N} \xi_i - \nu p$$
$$\text{s.t. } y_i \left(\mathcal{X}_i \prod_{j=1}^{M} \times_j \boldsymbol{w}_j + b \right) \ge p - \xi_i, \quad 1 \le i \le N$$
$$\xi_i \ge 0, \quad 1 \le i \le N \tag{8}$$

where $\boldsymbol{w}_j \in \mathbb{R}^{I_j}$, and $\mathcal{W} = \boldsymbol{w}_1 \circ, \ldots, \circ \boldsymbol{w}_m$ denotes the projecting vector along with the jth mode and the rank-1 projecting tensor. To solve the above optimization problem, alternating optimization technology is utilized, i.e., optimize each $\boldsymbol{w}_{i_m}$, in turn, by fixing the remaining $\boldsymbol{w}_i|_{i \ne i_m}$ for the kth iteration, where $i_m = \text{mod}(k, m) + 1$ denotes the index of the mode to be optimized. Thus, the optimization problem in the kth iteration is equivalent to the optimization problem of ν-SVM [see (5)]. Alternating optimization is continuously performed until the algorithm converges. Then, for the sample $\mathcal{X} \in \mathbb{R}^{I_1 \times \cdots \times I_M}$, the label can be predicted using the following decision function:

$$y(\mathcal{X}) = \text{sign}\left(\mathcal{X} \prod_{j=1}^{M} \times_j \boldsymbol{w}_j + b \right). \tag{9}$$

III. ADAPTIVE HSTM

In this section, to address input represented as heterogeneous tensors, the HSTM method is first proposed to use multitensor rank-R projecting instead of single-tensor rank-1 projecting for classical ν-STM to separate input in the heterogeneous tensor space. To recognize objects using arbitrary combinations of multisource heterogeneous data, the AHSTM is further developed to utilize shared projecting tensors learned from all possible combinations to effectively identify objects for arbitrary combination situations. A comparison of acceptable input data for different methods is presented in Table I.

A. Constructing HSTMs

As mentioned above, the STM takes the tensor as input and separates the data with different labels in tensor space, whereas it requires the input to be represented as a single tensor. With regard to the application for object recognition using multisource remote sensing data, the standard STM cannot directly process input that is naturally represented as heterogeneous tensors $\{\mathcal{X}^{(m)} | 1 \le m \le M\}$, where $\mathcal{X}^{(m)}$ denotes the mth tensor extracted from the object, and it has to permutate these tensors to a single tensor with a larger size.

Given the training set $\{\mathcal{X}^{(im)}|_{m=1}^{M}, y_i\}_{i=1}^{N}$, where $\mathcal{X}^{(im)}$ denotes the mth tensors for the ith sample, to process heterogeneous tensors effectively, instead of the rank-1 single rank-1 projecting tensor $\mathcal{W}$ in ν-STM [see (8)], HSTM constructs M rank-R projecting tensor for each input features tensor, i.e., $\{\mathcal{W}^{(m)} | \mathcal{W}^{(m)} = \sum_{r=1}^{R} \boldsymbol{w}_{(1\ r)}^{(m)} \circ, \ldots, \circ \boldsymbol{w}_{(l_m r)}^{(m)}, 1 \le m \le M\}$, which ensures that it can directly and accurately project input $\mathcal{X}^{(m)}|_{m=1}^{M}$ through $\mathcal{W}^{(m)}|_{m=1}^{M}$ to heterogeneous tensors space. Then, the resulting HSTM can be drawn as follows:

$$\min_{\boldsymbol{w}_{(lr)}^{(m)} \mid_{l=1,m=1,r=1}^{l_m,M,R}, b, p, \xi_i|_{i=1}^{N}} \frac{1}{2} \sum_{m=1}^{M} \sum_{r=1}^{R} \left\| \boldsymbol{w}_{(1\ r)}^{(m)} \circ, \ldots, \circ \boldsymbol{w}_{(l_m r)}^{(m)} \right\|_F^2$$
$$- \nu p + \frac{1}{N} \sum_{i=1}^{N} \xi_i$$
$$\text{s.t. } y_i \left(\sum_{m=1}^{M} \sum_{r=1}^{R} \mathcal{X}^{(im)} \prod_{l=1}^{l_m} \times_l \boldsymbol{w}_{(lr)}^{(m)} + b \right) \ge p - \xi_i$$
$$1 \le i \le N$$
$$\xi_i \ge 0, \quad 1 \le i \le N \tag{10}$$

where R, l_m, and $\mathcal{W}^{(m)} = \sum_{r=1}^{R} \boldsymbol{w}_{(1\ r)}^{(m)} \circ, \ldots, \circ \boldsymbol{w}_{(l_m r)}^{(m)}$ denote the largest CANDECOMP/PARAFAC (CP) rank [31] of the projecting tensor, the order of the mth tensor $\mathcal{X}^{(im)}$, and the mth rank-R projecting tensor, respectively.

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Similar to the solving of the STM, the alternating optimization scheme is taken into consideration to obtain the optimal parameters of the HSTM. In the kth iteration, optimize the projecting vector set $\{w|_{(i_m,r)}^{(m)}|1 \le r \le R, 1 \le m \le M, i_m = \mathrm{mod}(k, l_m) + 1\}$ by fixing the remaining projecting vectors, where mod$(\cdot)$ denotes the remainder operator. The kth optimization problem is shown in the following equation:

$$\begin{aligned}\min_{\boldsymbol{w}_{(i_m r)}^{(m)}|_{m=1,r=1}^{M,R},b,p} \quad & \frac{1}{2}\sum_{m=1}^{M}\sum_{r=1}^{R}\frac{1}{2\prod_{j\neq i_k}\boldsymbol{w}_{(jr)}^{(m)\,T}\boldsymbol{w}_{(jr)}^{(m)}}\left\|\boldsymbol{w}_{(i_m r)}^{(m)}\right\|_F^2 \\ & -\nu p+\frac{1}{N}\sum_{i=1}^{N}\xi_i \\ \text{s.t. } & y_i\left(\sum_{m=1}^{M}\sum_{r=1}^{R}\boldsymbol{x}_{(i,r)}^{(i_m m)\,T}\boldsymbol{w}_{(i_m r)}^{(m)}+b\right)\ge p-\xi_i \\ & \qquad 1\le i\le N \\ & \xi_i\ge 0,\quad 1\le i\le N \end{aligned} \tag{11}$$

where $\boldsymbol{x}_{(i,r)}^{(i_m m)} = \mathcal{X}^{(im)}\prod_{l\neq i_m}^{l_m}\times_l \boldsymbol{w}_{(lr)}^{(m)}$. Thus, the Lagrangian function for the optimization problem of (11) can be drawn as in the following equation:

$$\begin{aligned}&\mathcal{L}\left(\boldsymbol{w}_{(i_m r)}^{(m)}\Big|_{m=1,r=1}^{M,R},b,p,\xi_i|_{i=1}^{N},\boldsymbol{\alpha},\boldsymbol{\lambda}\right) \\ &\quad=\frac{1}{2}\sum_{m=1}^{M}\sum_{r=1}^{R}\left(\prod_{j\neq i_m}\boldsymbol{w}_{(jr)}^{(m)\,T}\boldsymbol{w}_{(jr)}^{(m)}\left\|\boldsymbol{w}_{(i_m r)}^{(m)}\right\|_F^2\right) \\ &\qquad-\nu p+\frac{1}{N}\times\sum_{j=1}^{N}\xi_j-\sum_{j=1}^{N}\lambda_j\xi_j \\ &\qquad-\sum_{i=1}^{N}\alpha_i\left[y_i\left(\sum_{m=1}^{M}\sum_{r=1}^{R}\boldsymbol{x}_{(i,r)}^{(i_m m)\,T}\boldsymbol{w}_{(i_m r)}^{(m)}+b\right)-p+\xi_i\right] \\ &\text{s.t. } \boldsymbol{\lambda}\ge 0,\quad \boldsymbol{\alpha}\ge 0\end{aligned} \tag{12}$$

where variables $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_N]$ and $\lambda = [\boldsymbol{\lambda}_1, \ldots, \lambda_N]$ denote the Lagrangian multipliers. Since the optimization problem of the kth iteration [see (11)] is convex QP, the Karush–Kuhn–Tucker (K.K.T.) conditions [32] are both necessary and sufficient conditions for the optimal solution. According to the K.K.T. conditions, i.e., the partial derivatives of the Lagrangian function with respect to the optimal $\boldsymbol{w}_{(i_m r)}^{(m)}$, b, ξ_i, and p are zeros, we have

$$\begin{aligned}&\frac{\partial\mathcal{L}}{\partial\boldsymbol{w}_{(i_m r)}^{(m)}}=0\Rightarrow \\ &\quad\boldsymbol{w}_{(i_m r)}^{(m)}=\frac{1}{\prod_{l\neq i_m}\boldsymbol{w}_{(lr)}^{(m)\,T}\boldsymbol{w}_{(lr)}^{(m)}}\sum_{i=1}^{N}\alpha_i y_i\boldsymbol{x}_{(i,r)}^{(i_m m)} \\ &\frac{\partial\mathcal{L}}{\partial b}=0\Rightarrow\sum_{i=1}^{N}y_i\alpha_i=0 \\ &\frac{\partial\mathcal{L}}{\partial\xi_i}=0\Rightarrow\frac{1}{N}-\lambda_i-\alpha_i=0 \\ &\frac{\partial\mathcal{L}}{\partial p}=0\Rightarrow\sum_{i=1}^{N}\alpha_i=\nu.\end{aligned} \tag{13}$$

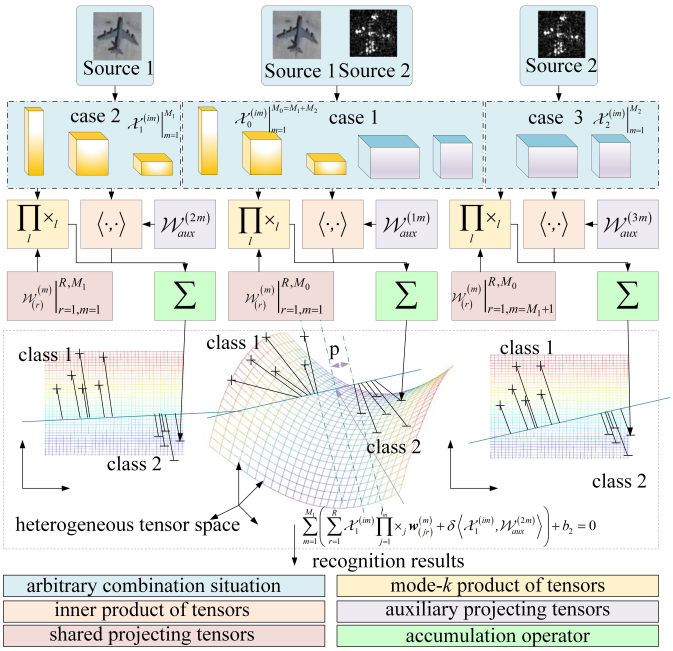


Fig. 2. Illustration of using AHSTM to address an arbitrary combination of multisource data.

Substituting the above expression into (12) yields the dual problem, as shown in the following equation:

$$\begin{aligned}\min_{\alpha}\ & \boldsymbol{\alpha}^T\left[\sum_{m=1}^{M}\sum_{r=1}^{R}\frac{1}{2\times\prod_{l\neq i_m}\boldsymbol{w}_{(lr)}^{(m)\,T}\boldsymbol{w}_{(lr)}^{(m)}}K_{m,r}\right]\boldsymbol{\alpha} \\ \text{s.t. } & \frac{1}{N}-\alpha_i\ge 0,\quad 1\le i\le N \\ & \alpha_i\ge 0,\quad 1\le i\le N \\ & \sum_i y_i\alpha_i=0 \\ & \sum_i\alpha_i=\nu\end{aligned} \tag{14}$$

where $K_{m,r}(i,j) = \langle y_i\boldsymbol{x}_{(i,r)}^{(i_m m)}, y_j\boldsymbol{x}_{(j,r)}^{(i_m m)}\rangle$. Equation (14) is equivalent to the dual optimization problem of the standard ν-SVM method [see (6)], which can be solved to obtain the optimal α using the corresponding ν-SVM solver. After solving (14), $\boldsymbol{w}_{(i_m r)}^{(m)}$ is updated as shown in (15), and the kth iteration is finished

$$\boldsymbol{w}_{(i_m r)}^{(m)}=\frac{1}{\prod_{l\neq i_m}\boldsymbol{w}_{(lr)}^{(m)\,T}\boldsymbol{w}_{(lr)}^{(m)}}\sum_{i=1}^{N}\alpha_i y_i\boldsymbol{x}_{(i,r)}^{(i_m m)}. \tag{15}$$

Repeat the iteration until the convergence criterion is satisfied, i.e., if the difference between the projecting vectors obtained in the $(k+1)$th iteration and those obtained in the kth iteration is less than a manually established tolerance or the number of iterations exceeds the preestablished maximum number of iterations [see (16)]

$$\sum_m\sum_R\sum_l\left\|\boldsymbol{w}_{(lr)}^{(m)}-\boldsymbol{w}_{(lr)}^{(m)'}\right\|_F^2<\delta_b||k>k_{\text{th}} \tag{16}$$

where $\boldsymbol{w}_{(lr)}^{(m)'}$, δ_b, and k_{th} denote the updated $\boldsymbol{w}_{(jr)}^{(m)}$ in the last iteration, with the tolerance established manually, and the

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Algorithm 1 Training Approach for HSTM

Input: training set $\left\{ \mathcal{X}^{(jm)} \middle| \begin{matrix} N, M \\ j=1, m=1 \end{matrix}, y_j \right\}$

Output: model parameters of HSTM, including $\mathbf{w}_{(lr)}^{(m)} \middle| \begin{matrix} l_m, R, M \\ l=1, r=1, m=1 \end{matrix}$, b and p.

Step 1: Initialize $k = 1$,and set random values to $\mathbf{w}_{(lr)}^{(m)} \middle| \begin{matrix} l_m, R, M \\ l=1, r=1, m=1 \end{matrix}$.

Step 2: Calculate $i_m = \text{mod}(k, l_m) + 1, 1 \le m \le M$, and select $\mathbf{w}_{(i_m r)}^{(m)} \Big|_{m=1,r=1}^{M,R}$ to be optimized by fixing other projecting vectors. Perform Eq. (12)-Eq. (15) to update $\mathbf{w}_{(i_m r)}^{(m)} \Big|_{m=1,r=1}^{M,R}$.

Step 3: If Eq. (16) is met, then calculate b and p, and output the model parameters; otherwise, let $k = k + 1$ and perform Step 2.

preestablished maximum number of iterations is used. After the iteration converges, using the K.K.T. conditions, we have

$$\begin{aligned}
&\sum_{j=1}^{N} \lambda_j \xi_j \ge 0 \\
&\sum_{i=1}^{N} \alpha_i \left[y_i \left(\sum_{m=1}^{M} \sum_{r=1}^{R} \mathcal{X}^{(im)} \prod_{l=1}^{l_m} \times_l \boldsymbol{w}_{(lr)}^{(m)} + b \right) - p + \xi_i \right] \ge 0 \\
&y_i \left(\sum_{m=1}^{M} \sum_{r=1}^{R} \boldsymbol{x}_{(i,r)}^{(i_m m)^T} \boldsymbol{w}_{(i_m r)}^{(m)} + b \right) - p + \xi_i \ge 0 \\
&\xi_j \ge 0 \\
&\frac{1}{N} - \lambda_i - \alpha_i = 0 \\
&\lambda_j \ge 0 \\
&\alpha_i \ge 0 \\
&\Rightarrow \\
&y_i \left(\sum_{m=1}^{M} \sum_{r=1}^{R} \mathcal{X}^{(im)} \prod_{l=1}^{l_m} \times_l \boldsymbol{w}_{(lr)}^{(m)} + b \right) - p = 0, \ 0 < \alpha_i < \frac{1}{N}.
\end{aligned} \tag{17}$$

Therefore, it can be considered to use the projecting values $\{\sum_{m=1}^{M} \sum_{r=1}^{R} \mathcal{X}^{(im)} \prod_{l=1}^{l_m} \times_l \boldsymbol{w}_{(lr)}^{(m)} | 0 < \alpha_i < \frac{1}{N}\}$ to calculate b and p as follows:

$$\begin{aligned}
b &= -0.5 \times \left(\frac{1}{\#\{i | y_i = 1\}} \times \sum_{\{i | y_i = 1\}} \sum_{m=1}^{M} \sum_{r=1}^{R} \mathcal{X}^{(im)} \prod_{l=1}^{l_m} \times_l \boldsymbol{w}_{(lr)}^{(m)} \right) \\
&\quad - 0.5 \times \left(\frac{1}{\#\{i | y_i = -1\}} \times \sum_{\{i | y_i = -1\}} \sum_{m=1}^{M} \sum_{r=1}^{R} \mathcal{X}^{(im)} \prod_{l=1}^{l_m} \times_l \boldsymbol{w}_{(lr)}^{(m)} \right) \\
p &= \frac{1}{\#\{i | y_i = 1\}} \times \sum_{\{i | y_i = 1\}} \sum_{m=1}^{M} \sum_{r=1}^{R} \mathcal{X}^{(im)} \prod_{l=1}^{l_m} \times_l \boldsymbol{w}_{(lr)}^{(m)} + b \\
&\qquad 0 < \alpha_i < \frac{1}{N}.
\end{aligned} \tag{18}$$

Finally, the training method for HSTM is finished, as summarized in Algorithm 1.

The predicted class label through HSTM can be calculated as (19) for any test sample

$$f_h \left(\mathcal{X}^{(m)} \big|_{m=1}^{M} \right) = \text{sign} \left(\sum_{m=1}^{M} \sum_{r=1}^{R} \mathcal{X}^{(m)} \prod_{l=1}^{l_m} \times_l \boldsymbol{w}_{(lr)}^{(m)} + b \right) \tag{19}$$

where h denotes the trained parameter set, that is,

$$h = \left\{ \boldsymbol{w}_{(lr)}^{(m)} \middle| \begin{matrix} l_m, R, M \\ l=1, r=1, m=1 \end{matrix}, b, p \right\}.$$

B. Extending to the AHSTM

Although the HSTM can accept input represented as heterogeneous tensors, it is not suitable for addressing arbitrary combinations of multisource data. For arbitrary combination situations, the HSTM and other existing classification algorithms must train the classifier separately for each case. It will consume extra storage space to record multiple classifier parameters and lose the related information between different combinations, which would lead to decreased classification performance.

To overcome the problem that the existing classification algorithms are not suitable for arbitrary combinations, the AHSTM is proposed to effectively process arbitrary combinations of heterogeneous multisource data, as illustrated in Fig. 2. Suppose that the number of sources is equal to 2 (see Fig. 1); $\mathcal{X}_1^{(im)}|_{m=1}^{M_1}$ and $\mathcal{X}_2^{(im)}|_{m=1}^{M_2}$ denote the heterogeneous tensors extracted from source 1 and source 2, respectively, and the samples extracted from cases 1, 2, and 3 are denoted as $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2}\}$, $\mathcal{X}_1^{(im)}|_{m=1}^{M_1}$, and $\mathcal{X}_2^{(im)}|_{m=1}^{M_2}$, respectively.

Before constructing AHSTM, it is necessary to consider whether HSTM with a single set of parameters can be used to address different cases rather than train different HSTM models for each case. In this way, the definition of the generalization error is introduced, and Proposition 1 is given, as follows:

Definition 4 (Generalization Error): The generalization error is a measure of how accurately an algorithm can predict the labels of previously unseen samples that obey the specific distribution $\mathcal{D}_1$, that is,

$$E(h_2; \mathcal{D}_1) = P_{\mathcal{X}_1^{(im)}|_{M_1}^{m=1} \sim \mathcal{D}_1} \left(f_{h_2} \left(\mathcal{X}_1^{(im)} \Big|_{m=1}^{M_1} \right) \ne y_i \right) \tag{20}$$

where $E(\cdot)$, h_2, $f_{h_2}(\cdot)$, $\mathcal{D}_1$, and $P(\cdot)$ denote the generalization error, parameters of the model, training classifier, distribution of samples, and probability, respectively.

Proposition 1: Considering the general setting in machine learning, the extracted heterogeneous tensors from the same source obey the same distribution, i.e., $\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \sim \mathcal{D}_1$ and $\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_2} \sim \mathcal{D}_2$ for different cases. Then, suppose that the generalization error of HSTM with parameters $h_2 = \left\{ \boldsymbol{w}_{(2\ rj)}^{(m)} \middle| \begin{matrix} l_m, R, M_1 \\ j=1, r=1, m=1 \end{matrix}, b_2 \right\}$ for case 2 and HSTM

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with parameters $h_3 = \left\{ \boldsymbol{w}^{(m)}_{(3\ rj)} \middle| \begin{matrix} l_{m+M_1}, R, M_2 \\ j=1, r=1, m=1 \end{matrix}, b_3 \right\}$ for case 3 are less than ε_1 and ε_2, respectively. In other words, we have

$$E(h_2; \mathcal{D}_1) = P_{\mathcal{X}_1^{(im)}|_{m=1}^{M_1} \sim \mathcal{D}_1} \left(f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \right) < \varepsilon_1$$
$$E(h_3; \mathcal{D}_2) = P_{\mathcal{X}_2^{(im)}|_{m=1}^{M_2} \sim \mathcal{D}_2} \left(f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_2} \right) \neq y_i \right) < \varepsilon_2. \tag{21}$$

Then, there exists an HSTM with parameters

$$h_1 = \left\{ \boldsymbol{w}^{(m)}_{(2\ rj)} \middle| \begin{matrix} l_m, R, M_1 \\ j=1, r=1, m=1 \end{matrix}, \boldsymbol{w}^{(m)}_{(3\ rj)} \middle| \begin{matrix} l_{m+M_1}, R, M_2 \\ j=1, r=1, m=1 \end{matrix}, b_1 \right\}$$

whose generalization error for case 1 is less than $\varepsilon_1+\varepsilon_2$, which is

$$E(h_1; \mathcal{D}) = P_{\left\{ \mathcal{X}_1^{(im)}|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \left(f_{h_1}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}\Big|_{m=1}^{M_2} \right) \neq y_i \right) < \varepsilon_1 + \varepsilon_2. \tag{22}$$

Proof: For heterogeneous tensors $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}\}$ and $\{\mathcal{X}_2^{(im)}|_{m=1}^{M_2}\}$ from case 1, suppose that $\mathcal{X}_1^{(im)}|_{m=1}^{M_1}$ and $\mathcal{X}_2^{(im)}|_{m=1}^{M_2}$ can be recognized by HSTM with parameters h_2 and h_3, respectively, i.e., (23) is satisfied

$$y_i \left(\sum_{m=1}^{M_1} \sum_{r=1}^{R} \mathcal{X}_1^{(im)} \prod_{j=1}^{l_m} \times_j \boldsymbol{w}^{(m)}_{(2\ rj)} + b_2 \right) > 0$$
$$y_i \left(\sum_{m=1}^{M_2} \sum_{r=1}^{R} \mathcal{X}_2^{(im)} \prod_{j=1}^{l_m} \times_j \boldsymbol{w}^{(m)}_{(3rj)} + b_3 \right) > 0. \tag{23}$$

Adding the above two expressions, we have

$$y_i \left(\sum_{m=1}^{M_1} \sum_{r=1}^{R} \mathcal{X}_1^{(im)} \prod_{j=1}^{l_m} \times_j \boldsymbol{w}^{(m)}_{(2\ rj)} + \sum_{m=1}^{M_2} \sum_{r=1}^{R} \mathcal{X}_2^{(im)} \prod_{j=1}^{l_m} \times_j \boldsymbol{w}^{(m)}_{(3rj)} + b_2 + b_3 \right) > 0. \tag{24}$$

Therefore, there exists an HSTM with parameters

$$h_1 = \left\{ \boldsymbol{w}^{(m)}_{(2\ rj)} \middle| \begin{matrix} l_m, R, M_1 \\ j=1, r=1, m=1 \end{matrix}, \boldsymbol{w}^{(m)}_{(3\ rj)} \middle| \begin{matrix} l_{m+M_1}, R, M_2 \\ j=1, r=1, m=1 \end{matrix}, b_2 + b_3 \right\}$$

that can be used to recognize the heterogeneous tensors $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2}\}$ from case 1. In other words, we have

$$P_{\left\{ \mathcal{X}_1^{(im)}|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \times \left(\frac{f_{h_1}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i}{f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) = y_i,\ f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) = y_i} \right) = 0. \tag{25}$$

This means that, if the single-source samples $\mathcal{X}_1^{(im)}|_{m=1}^{M_1}$ and $\mathcal{X}_2^{(im)}|_{m=1}^{M_2}$ split from the multisource sample $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2}\}$ and can be recognized by HSTM with parameters h_2 and h_3, respectively, then the corresponding multisource sample $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2}\}$ must be recognized by HSTM with parameter h_1. Then, for the generalization error of HSTM with parameters h_1, applying the basic total probability theorem to $E(h_1; \mathcal{D})$, we obtain

$$\begin{aligned} &E(h_1; \mathcal{D}) \\ &= P_{\left\{ \mathcal{X}_1^{(im)}|_{m=11}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \\ &\quad \times \left(\frac{f_{h_1}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i}{f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) = y_i,\ f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) = y_i} \right) \\ &\quad \times P_{\left\{ \mathcal{X}_1^{(im)}|_{m=11}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \left(f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) = y_i,\ f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) = y_i \right) \\ &\quad + P_{\left\{ \mathcal{X}_1^{(im)}|_{m=11}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \\ &\quad \times \left(\frac{f_{h_1}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i}{f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \,||\, f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i} \right) \\ &\quad \times P_{\left\{ \mathcal{X}_1^{(im)}|_{m=11}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \\ &\quad \times \left(f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \,||\, f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \right). \end{aligned} \tag{26}$$

Substituting (25) into (26) and then using basic aggregate theory and the known condition in (21), we have

$$\begin{aligned} &E(h_1; \mathcal{D}) \\ &= P_{\left\{ \mathcal{X}_1^{(im)}|_{m=11}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \\ &\quad \times \left(\frac{f_{h_1}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i}{f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \,||\, f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i} \right) \\ &\quad \times P_{\left\{ \mathcal{X}_1^{(im)}|_{m=11}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \\ &\quad \times \left(f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \,||\, f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \right) \\ &\leq P_{\left\{ \mathcal{X}_1^{(im)}|_{m=11}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \\ &\quad \times \left(f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \,||\, f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \right) \\ &\leq P_{\left\{ \mathcal{X}_1^{(im)}|_{m=11}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \left(f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \right) \\ &\quad + P_{\left\{ \mathcal{X}_1^{(im)}|_{m=11}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2} \right\} \sim \mathcal{D}} \left(f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \right) \\ &= P_{\mathcal{X}_1^{(im)}|_{m=1}^{M_1} \sim \mathcal{D}_1} \left(f_{h_2}\left(\mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1} \right) \neq y_i \right) \end{aligned}$$

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$$+ P_{\mathcal{X}_2^{(im)}|_{m=1}^{M_1} \sim \mathcal{D}_2}\left(f_{h_3}\left(\mathcal{X}_2^{(im)}\Big|_{m=1}^{M_1}\right) \neq y_i\right) \leq \varepsilon_1 + \varepsilon_2. \tag{27}$$

The proof is completed.

It is known from Proposition 1 that, if the generalization error for HSTM with h_2 (or h_3) for case 2 (or case 3) is small enough, the HSTM with h_1 can also obtain a sufficiently small generalization error for case 1, where h_1 has the same projecting tensors as h_2 and h_3. This finding indicates that it is feasible to extend HSTM to handle arbitrary combinations using a single set of parameters. We are given a training set $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2}, y_i\}|_{i=1}^{N_1}$ from case 1, $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, y_i\}|_{i=1}^{N_2}$ from case 2, and $\{\mathcal{X}_2^{(im)}|_{m=1}^{M_2}, y_i\}|_{i=1}^{N_3}$ from case 3. It is considered to jointly learn the same projecting tensors $\{\mathcal{W}_{(r)}^{(m)}|\boldsymbol{w}_{(lr)}^{(m)}\circ, \ldots, \circ\boldsymbol{w}_{(l_m r)}^{(m)}, 1 \leq r \leq R, 1 \leq m \leq M_0\}$ according to samples from three cases in such a way that the resulting classifier can adapt to arbitrary combinations. In this way, the optimization problem of the proposed AHSTM can be drawn as follows:

$$\begin{aligned}
\min_{\boldsymbol{w}_{(lr)}^{(m)}\big|_{l=1,r,m=1}^{l_m,R,M_0}, \mathcal{W}_{\text{aux}}^{(jm)}\big|_{j=1,m}^{c}, b, p} \; & \frac{1}{2}\left(\sum_{m=1}^{M_1+M_2}\sum_{r=1}^{R}\left\|\boldsymbol{w}_{(lr)}^{(m)}\circ, \ldots, \circ\boldsymbol{w}_{(l_m r)}^{(m)}\right\|_F^2 + \sum_{j=1}^{N_c}\sum_{m=1}^{M_{j-1}}\left\|\mathcal{W}_{\text{aux}}^{(jm)}\right\|_F^2\right) \\
& + \sum_{j=1}^{c}\frac{1}{N_j}\sum_{i=1}^{N_j}\xi_{ji} - \sum_{j=1}^{c}\nu_j p_j \\
\text{s.t. } & y_{ji}\left(\sum_{m=1}^{M_{j-1}}\left(\left(\sum_{r=1}^{R}\mathcal{X}_{j-1}^{(im)}\prod_{l=1}^{l_m}\times_l \boldsymbol{w}_{(lr)}^{\left(m+M_j'\right)}\right) + \delta\left\langle\mathcal{X}_{j-1}^{(im)}, \mathcal{W}_{\text{aux}}^{(jm)}\right\rangle\right) + b_j\right) \\
& \geq p_j - \xi_{ji}, \quad \xi_{ji} > 0, \; 1 \leq i \leq N_j, \; 1 \leq j \leq c
\end{aligned} \tag{28}$$

where b_j, ν_j, p_j $\xi_{1\,i}$, $\mathcal{W}_{\text{aux}}^{(jm)}$, δ, N_j, and $\left\{\boldsymbol{w}_{(i_m r)}^{(m)}\middle| \begin{matrix} R, M_0 \\ r=1, m=1 \end{matrix}\right\}$ denote the basis of case j, the ν parameter ($0 \leq \nu_j \leq 1$) of case j, the projecting margin between the support tensors and classification hyperplane for case j, the slack variable for the ith sample from case 1, the mth auxiliary projecting tensor for case j, the weight of the auxiliary projecting tensors, the number of training samples for case j, and the shared projecting vectors. N_c denotes the number of possible combinations for multisource data, and it is set to 3 for two-source object recognition in this article. Then, $M_j' = \begin{cases} 0, & \text{if } j \in \{1,2\} \\ \sum_{h=1}^{j-2} M_h, & \text{if } j \geq 3 \end{cases}$ denotes the correspondence between shared projecting vectors and multisource samples. For the convenience of writing, $\mathcal{X}_0^{(im)}|_{m=1}^{M_0=M_1+M_2} = \{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2}\}$ is used to represent the extracted heterogeneous tensors for the ith sample from case 1. Since the samples from case 1 contain richer information than those from cases 2 and 3, the projecting margin of case 1 should be larger than those of cases 2 and 3. Thus, p_j are assigned to each combination to adjust the proper projecting margin adaptively in AHSTM.

In AHSTM, $\{\mathcal{W}_{(r)}^{(m)}|\boldsymbol{w}_{(lr)}^{(m)}\circ, \ldots, \circ\boldsymbol{w}_{(l_m r)}^{(m)}, 1 \leq r \leq R, \; 1 \leq m \leq M_0\}$ is defined as shared projecting tensors learned jointly from samples of three cases and is used to describe the related information between different combinations. In addition, considering that there exists individual information in a specific combination, the auxiliary projecting tensors $\mathcal{W}_{\text{aux}}^{(jm)}|_{j,m}$ learned separately from each combination are constructed to mine the individual information in each combination. When δ is equal to 0, the auxiliary projecting tensors can be ignored. When δ is too large, the shared projecting tensors are negligible, and the AHSTM is degenerated to ν-SVM trained separately for each combination.

To solve the optimization problem of AHSTM, similar to solving HSTM, the alternating optimization scheme is utilized to obtain the optimal solution. For the kth iteration, we optimize the shared projecting vectors $\left\{\boldsymbol{w}_{(i_m r)}^{(m)}\middle| \begin{matrix} R, M_1+M_2 \\ r=1, m=1 \end{matrix}\right\}$ and all of the auxiliary projecting tensors $\mathcal{W}_{\text{aux}}^{(jm)}|_{j,m}$ by fixing other shared projecting vectors, where $i_m = \text{mod}(k, l_m) + 1, 1 \leq m \leq M_0$. Then, a Lagrangian function is constructed to eliminate the constraints, as follows:

$$\begin{aligned}
& \mathcal{L}\left(\boldsymbol{w}_{(i_m r)}^{(m)}\middle| \begin{matrix} R, M_1+M_2 \\ r=1, m=1 \end{matrix}, \mathcal{W}_{\text{aux}}^{(j,m)}\big|_{j=1,m}^{3}\boldsymbol{b}, \boldsymbol{p}, \xi_{ji}\big|_{j=1,i=1}^{3,N}, \boldsymbol{\alpha}_j\big|_{j=1}^{3}, \boldsymbol{\lambda}_j\big|_{j=1}^{3}\right) \\
& = \frac{1}{2}\sum_{m=1}^{M_1+M_2}\left(\sum_{r=1}^{R}\left\|\boldsymbol{w}_{(lr)}^{(m)}\circ, \ldots, \circ\boldsymbol{w}_{(l_m r)}^{(m)}\right\|_F^2 + \sum_{j=1}^{3}\sum_{m=1}^{M_{j-1}}\left\|\mathcal{W}_{\text{aux}}^{(jm)}\right\|_F^2\right) \\
& \quad + \sum_{j=1}^{3}\frac{1}{N_j}\sum_{i=1}^{N_j}\xi_{ji} - \sum_{j=1}^{3}\nu_j p_j - \sum_{j=1}^{3}\sum_{i=1}^{N_i}\lambda_{ji}\xi_{ji} \\
& \quad - \sum_{j=1}^{3}\sum_{i=1}^{N_j}\alpha_{ji}\left[y_{ji}\left(\sum_{m=1}^{M_{j-1}}\left(\left(\sum_{r=1}^{R}\mathcal{X}_{j-1}^{(im)}\prod_{l=1}^{l_m}\times_l \boldsymbol{w}_{(lr)}^{\left(m+M_j'\right)}\right) + \delta\left\langle\mathcal{X}_{j-1}^{(im)}, \mathcal{W}_{\text{aux}}^{(j,m)}\right\rangle\right) + b_j\right) - p_j + \xi_{ji}\right]
\end{aligned} \tag{29}$$

where $\boldsymbol{\lambda}_j = [\lambda_{j1}, \ldots, \lambda_{jN_j}]$ and $\boldsymbol{\alpha}_j = [\alpha_{j1}, \ldots, \alpha_{jN_j}]$ denote the dual variables. Let the partial derivatives of the above formula with respect to variables be zero; then, we have

$$\begin{aligned}
& \frac{\partial\mathcal{L}}{\partial\boldsymbol{w}_{(i_m r)}^{(m)}} = 0 \Rightarrow \\
& \boldsymbol{w}_{(i_m r)}^{(m)} = \frac{1}{\prod_{l\neq i_m}\left\|\boldsymbol{w}_{(lr)}^{(m)}\right\|_F^2}\sum_{j=1}^{3}\sum_{i=1}^{N_j}\alpha_{ji}y_{ji}\mathcal{X}_{j-1}^{(im)}\prod_{l\neq i_m}\times_l \boldsymbol{w}_{(lr)}^{\left(m+M_j'\right)} \\
& \frac{\partial\mathcal{L}}{\partial\mathcal{W}_{\text{aux}}^{(jm)}} = 0 \Rightarrow
\end{aligned}$$

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Algorithm 2 Training Approach for AHSTM

Input: sample sets of different combinations, including case 1 $\left\{ \mathcal{X}_0^{(im)}\Big|_{m=1}^{M_1+M_2}, y_{1i} \right\}\Big|_{i=1}^{N_1}$, case 2 $\left\{ \mathcal{X}_1^{(im)}\Big|_{m=1}^{M_1}, y_{2i} \right\}\Big|_{i=1}^{N_2}$ and case 3 $\left\{ \mathcal{X}_2^{(im)}\Big|_{m=M_1+1}^{M_2}, y_{3i} \right\}\Big|_{i=1}^{N_3}$

Output: The parameters for AHSTM, i.e., $\mathbf{w}_{(lr)}^{(m)}\Big| \begin{matrix} l_m, R, M_1+M_2 \\ l=1, r=1, m=1 \end{matrix}$, $\mathcal{W}_{\text{aux}}^{(jm)}\big|_{j,m}$, $\boldsymbol{b}$, $\boldsymbol{p}$

Step 1: Initialize $\mathbf{w}_{(lr)}^{(m)}\Big| \begin{matrix} l_m, R, M_1+M_2 \\ l=1, r=1, m=1 \end{matrix}$, $\mathcal{W}_{\text{aux}}^{(jm)}\big|_{j,m}$ by a random number generator, and let $k=1$.

Step 2: Let $i_m = \text{mod}(k, l_m) + 1$, and optimize $\mathbf{w}_{(i_m r)}^{(m)}\Big| \begin{matrix} M_1+M_2, R \\ m=1, r=1 \end{matrix}$ and $\mathcal{W}_{\text{aux}}^{(jm)}\big|_{j,m}$ by fixing other shared projecting vectors according to Eq. (29)-Eq. (34).

Step 3: If the convergence condition in Eq. (35) is met, calculate b_j and p_j by Eq. (36); otherwise, let $k=k+1$ and perform Step 2.

$$
\begin{aligned}
\mathcal{W}_{\text{aux}}^{(jm)} &= \delta \sum_{i=1}^{N_j} \alpha_{ji} y_{ji} \mathcal{X}_{j-1}^{(im)} \\
\frac{\partial \mathcal{L}}{\partial b_j} &= 0 \Rightarrow \\
\sum_{i=1}^{N_j} y_{ji} \alpha_{ji} &= 0 \\
\frac{\partial \mathcal{L}}{\partial \xi_{ji}} &= 0 \Rightarrow \\
\frac{1}{N_j} - \lambda_{ji} - \alpha_{ji} &= 0 \\
\frac{\partial \mathcal{L}}{\partial p_j} &= 0 \Rightarrow \\
\sum_{i=1}^{N_j} \alpha_{ji} &= \nu_j .
\end{aligned} \tag{30}
$$

Substituting the above formula into the Lagrangian function, we obtain the dual problem, as follows:

$$
\begin{aligned}
&\min_{\alpha} \boldsymbol{\alpha}^T Q \boldsymbol{\alpha} \\
&\text{s.t. } \frac{1}{N_j} - \alpha_{ji} \geq 0, \quad 1 \leq i \leq N_j, \ 1 \leq j \leq 3 \\
&\quad \alpha_{ji} \geq 0, \quad 1 \leq i \leq N_j, \ 1 \leq j \leq 3 \\
&\quad \sum_i y_{ji} \alpha_{ji} = 0, \quad 1 \leq j \leq 3 \\
&\quad \sum_i \alpha_{ji} = \nu_j, \quad 1 \leq j \leq 3
\end{aligned} \tag{31}
$$

where $\boldsymbol{\alpha} = [\alpha_{11}, \ldots, \alpha_{1N_1}, \ldots, \alpha_{31}, \ldots, \alpha_{3N_3}]^T$, $Q = \sum_{m=1}^{M_1+M_2} \sum_{r=1}^{R} K_{m,r}$, and superscript T denotes the transposition operation. The value for entry $(\sum_{j=1}^{g_1-1} N_j + h_1, \sum_{j=1}^{g_2-1} N_j + h_2)$ in $K_{m,r}$ is shown in the following equation:

$$
\begin{aligned}
&K_{m,r}\left(\sum_{j=1}^{g_1-1} N_j + h_1, \sum_{j=1}^{g_2-1} N_j + h_2 \right) \\
&= \frac{1}{2 \prod_{l \neq i_m} \left\| \boldsymbol{w}_{(lr)}^{(m)} \right\|_F^2} \\
&\quad \times \left\langle y_{g_1 h_1} \mathcal{X}_{g_1-1}^{(h_1 m)} \prod_{l \neq i_m} \times_l \boldsymbol{w}_{(lr)}^{\left(m+M'_{g_1}\right)}, \right. \\
&\qquad \left. y_{g_2 h_2} \mathcal{X}_{g_2-1}^{(h_2 m)} \prod_{l \neq i_m} \times_l \boldsymbol{w}_{(lr)}^{\left(m+M'_{g_2}\right)} \right\rangle \\
&\quad + \frac{\delta^2}{2R} \times c(g_1, g_2) \times \left\langle y_{g_1 h_1} \mathcal{X}_{g_1-1}^{(h_1 m)}, y_{g_2 h_2} \mathcal{X}_{g_2-1}^{(h_2 m)} \right\rangle
\end{aligned} \tag{32}
$$

where $c(g_1, g_2) = \begin{cases} 1, & \text{if } g_1 = g_2 \\ 0, & \text{otherwise.} \end{cases}$ When calculating (32), the value of $K_{m,r}(\sum_{j=1}^{g_1-1} N_j + h_1, \sum_{j=1}^{g_2-1} N_j + h_2)$ will be set to zero if $m > \min\{M_{g_1-1}, M_{g_2-1}\}$. After using the QP solving method (a more effective solving method is presented in Section IV) to solve the dual problem in (31), according to the stationarity of the K.K.T. conditions in (30), the resulting $\boldsymbol{\alpha}$ is used to update $\boldsymbol{w}_{(i_m r)}^{(m)}$ and $\mathcal{W}_{\text{aux}}^{(jm)}$, as seen in the following equations:

$$
\boldsymbol{w}_{(i_m r)}^{(m)} = \frac{1}{\prod_{l \neq i_m} \left\| \boldsymbol{w}_{(lr)}^{(m)} \right\|_F^2} \sum_{j=1}^{3} \sum_{i=1}^{N_j} \alpha_{ji} y_{ji} \mathcal{X}_j^{(im)} \prod_{l \neq i_m} \times_l \boldsymbol{w}_{(lr)}^{\left(m+M'_j\right)} \tag{33}
$$

$$
\mathcal{W}_{\text{aux}}^{(jm)} = \delta \times \sum_{i=1}^{N_j} \alpha_{ji} y_{ji} \mathcal{X}_j^{(im)}. \tag{34}
$$

Similar to HSTM, the convergence criterion of AHSTM for alternating optimization is given in the following equation:

$$
\left\{ \sum_m \sum_R \sum_l \left\| \boldsymbol{w}_{(lr)}^{(m)} - \boldsymbol{w}_{(lr)}^{(m)'} \right\|_F^2 + \sum_m \sum_j \left\| \mathcal{W}_{\text{aux}}^{(jm)} - \mathcal{W}_{\text{aux}}^{(jm)'} \right\|_F^2 < \delta_g \right\} \quad k_{\text{iter}} > k_{\text{th}} \tag{35}
$$

where δ_g and k_{th} denote the preestablished tolerance and the maximum number of iterations, respectively. $\boldsymbol{w}_{(lr)}^{(m)'}$ ($\mathcal{W}_{\text{aux}}^{(jm)'}$) denotes the optimal $\boldsymbol{w}_{(lr)}^{(m)}$ ($\mathcal{W}_{\text{aux}}^{(jm)}$) in the last iteration. If the convergence conditions are not met, let $k=k+1$, and perform the next iteration; otherwise, stop the iterations and calculate parameters $\boldsymbol{b}$ and $\boldsymbol{p}$ by the following equation:

$$
\begin{aligned}
b_j = &-0.5 \\
&\times \left(\frac{1}{\#\left\{ k | y_{jk} = 1, 0 < \alpha_{jk} < \frac{1}{N_j} \right\}} \right. \\
&\quad \times \left. \sum_{\left\{ k | y_{jk}=1, 0<\alpha_{jk}<\frac{1}{N_j} \right\}} \sum_{m=1}^{M_{j-1}} \left(\sum_{r=1}^{R} \mathcal{X}_{j-1}^{(km)} \prod_{i=1}^{l_m} \times_i \boldsymbol{w}_{(ir)}^{\left(m+M'_j\right)} \right) \right)
\end{aligned}
$$

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$$+\delta\left\langle \mathcal{X}_{j-1}^{(km)}, \mathcal{W}_{\text{aux}}^{(jm)} \right\rangle \Bigg) - 0.5$$

$$\times \Bigg(\frac{1}{\#\left\{k|y_{jk}=-1, 0<\alpha_{jk}<\frac{1}{N_j}\right\}} \times \sum_{\left\{k|y_{jk}=-1,0<\alpha_{jk}<\frac{1}{N_j}\right\}} \sum_{m=1}^{M_{j-1}} \left(\sum_{r=1}^{R} \mathcal{X}_{j-1}^{(km)} \prod_{i=1}^{l_m} \times_i \boldsymbol{w}_{(ir)}^{\left(m+M_j'\right)} \right) + \delta\left\langle \mathcal{X}_{j-1}^{(km)}, \mathcal{W}_{\text{aux}}^{(jm)} \right\rangle \Bigg)$$

$$p_j = \Bigg(\frac{1}{\#\left\{k|y_{jk}=1, 0<\alpha_{jk}<\frac{1}{N_j}\right\}} \times \sum_{\left\{k|y_{jk}=1,0<\alpha_{jk}<\frac{1}{N_j}\right\}} \sum_{m=1}^{M_{j-1}} \left(\sum_{r=1}^{R} \mathcal{X}_{j-1}^{(km)} \prod_{i=1}^{l_m} \times_i \boldsymbol{w}_{(ir)}^{\left(m+M_j'\right)} \right) + \delta\left\langle \mathcal{X}_{j-1}^{(km)}, \mathcal{W}_{\text{aux}}^{(jm)} \right\rangle \Bigg) + b_j. \tag{36}$$

Here, the training for AHSTM is finished, and the training process is summarized in Algorithm 2.

According to the trained parameters, the class label of test sample $\mathcal{X}_{j-1}^{(m)}|_m$ from case j is predicted as follows:

$$y\left(\mathcal{X}_{j-1}^{(m)}|_m\right) = \text{sign}\left(\sum_m \left(\left(\sum_{r=1}^{R} \mathcal{X}_{j-1}^{(m)} \prod_{l=1}^{l_m} \times_l \boldsymbol{w}_{(lr)}^{\left(m+M_j'\right)} \right) + \delta\left\langle \mathcal{X}_{j-1}^{(m)}, \mathcal{W}_{\text{aux}}^{(jm)} \right\rangle \right) + b_j \right). \tag{37}$$

Our work focuses on object recognition using an arbitrary combination of two-source data. In fact, it is not difficult to generalize the optimization problem of AHSTM [see (28)] to classify objects using an arbitrary combination of S-source data, where S denotes any positive integer. For this situation, there are $2^S - 1$ possible combinations, and the parameter N_c in the optimization problem of AHSTM [see (28)] should be set to $2^S - 1$ to address arbitrary combinations of S-source data. The detailed solving procedure is similar to the above, with no more tautology here.

IV. Training AHSTM Using a Decomposition Algorithm

According to the training approach of AHSTM, it is necessary to solve the dual problem of (31) in each iteration of alternating optimization. Note that the dual problem of (31) is different from that of ν-SVM [see (6)]. Therefore, the dual problem of AHSTM cannot be solved using efficient SVM solvers and can only be solved using general optimization algorithms (e.g., [35] and [36]). General optimization algorithms, e.g., the interior-point method [35], will increase the space complexity because all the variables are updated in each iteration simultaneously. In addition, the size of the dual problem of AHSTM scales quadratically in the size of the training set, making it challenging to handle extensive training sets using the typical optimization solving method. To solve the dual problem of AHSTM efficiently, similar to the classic sequential minimal optimization (SMO) algorithm [33] and decomposition algorithm [34] used to solve C-SVM and ν-SVM, respectively, the dual problem of the AHSTM oriented decomposition algorithm is proposed to break the complex numerical QP problem into a series of simple analytic QP problems, which ensures efficiency for training AHSTM.

Since the dual problem of HSTM can be considered to be a particular situation from that of AHSTM, i.e., the number of combinations is equal to 1, for convenience and without loss of generality, the dual problem of AHSTM [see (31)] is taken as an example to describe the proposed decomposition method.

According to the constraints in (31), it is easy to construct the initial feasible solution by evenly splitting the value of ν_j to each variable α_{ji}, as shown in the following equation:

$$\alpha_{ji} = \frac{\nu_j}{2 \times \#\left|\left\{y_{jk}|y_{jk}=y_{ji}, 1\le k\le N_j\right\}\right|}. \tag{38}$$

Thus, to avoid updating all of the variables simultaneously, an iteration optimization process is established. In each iteration, the variables are split into the active part and the inactive part, denoted as the working set and nonworking set, respectively; only variables in the working set are updated, and the remaining variables are fixed. In detail, for the kth iteration, some variables in $\boldsymbol{\alpha}_{k_{\text{iter}}} = [\alpha_{k_{\text{iter}}1}, \ldots, \alpha_{k_{\text{iter}}N_{k_{\text{iter}}}}]$ are selected as working set $\boldsymbol{\alpha}_{k_{\text{iter}}B}$, the remaining variables in $\boldsymbol{\alpha}$ are denoted as nonworking set $\boldsymbol{\alpha}_N$, where $k_{\text{iter}} = \text{mod}(k, c) + 1$, and $c = 3$ denotes the number of combinations. Then, the dual problem in the kth iteration is rewritten from (31) as follows:

$$\begin{aligned} \min_{\boldsymbol{\alpha}_{k_{\text{iter}}B}} & f\left(\boldsymbol{\alpha}_{k_{\text{iter}}B}\right) = \boldsymbol{\alpha}_{k_{\text{iter}}B}^T Q_{BB} \boldsymbol{\alpha}_{k_{\text{iter}}B} + 2 \times \boldsymbol{\alpha}_{k_{\text{iter}}B}^T Q_{BN} \boldsymbol{\alpha}_N \\ \text{s.t. } & \boldsymbol{y}_{k_{\text{iter}}B}^T \boldsymbol{\alpha}_{k_{\text{iter}}B} = -\boldsymbol{y}_{k_{\text{iter}}N}^T \boldsymbol{\alpha}_{k_{\text{iter}}N} \\ & \boldsymbol{e}_{k_{\text{iter}}B}^T \boldsymbol{\alpha}_{k_{\text{iter}}B} = \nu_{k_{\text{iter}}} - \boldsymbol{e}_{k_{\text{iter}}N}^T \boldsymbol{\alpha}_{k_{\text{iter}}N} \\ & 0 \le \boldsymbol{\alpha}_{k_{\text{iter}}B}(i) \le \frac{1}{N_{k_{\text{iter}}}}, \quad 1 \le i \le N_{k_{\text{iter}}} \end{aligned} \tag{39}$$

where $\boldsymbol{\alpha}_{k_{\text{iter}}N} = \{\boldsymbol{\alpha}_{k_{\text{iter}}}\}/\boldsymbol{\alpha}_{k_{\text{iter}}B}, \boldsymbol{e}_{jB} = [1, \ldots, 1]$. $\begin{bmatrix} Q_{BB}, Q_{BN} \\ Q_{BN}^T, Q_{NN} \end{bmatrix}$ is the permutation of matrix Q according to the index of the working set and nonworking set. Similar to the selection strategy of the working set in [34], the feasible steepest descent direction of $\boldsymbol{\alpha}_{k_{\text{iter}}}$ for (39) is used to determine our working set, as shown in the following equation:

$$\begin{aligned} \min_{\boldsymbol{d}} & \nabla f\left(\boldsymbol{\alpha}_{k_{\text{iter}}}\right)^T \boldsymbol{d} \\ \text{s.t. } & \sum_i y_{k_{\text{iter}}i} d_i = 0 \\ & \sum_i d_i = 0 \end{aligned}$$

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Algorithm 3 Solving the Dual Problem of AHSTM Using the Proposed Decomposition Algorithm

Input: the Q of the dual problem of AHSTM
Output: optimal $\boldsymbol{\alpha}$

Step 1: Initialize $\boldsymbol{\alpha}$ using Eq. (38), and let $k = 1$.
Step 2: Calculate $k_{\text{iter}} = \text{mod}(k, c) + 1$, and select the working set using Eq. (41).
Step 3: Calculate step-size d_{step} using Eq. (42)-Eq. (46), and update the variables in the working set by Eq. (47).
Step 4: If the convergence condition in Eq. (48) is met, output the current $\boldsymbol{\alpha}$; otherwise, let $k = k + 1$ and perform Step 2.

$$\begin{aligned} & d_i \neq -1, \quad \text{if } \boldsymbol{\alpha}_{k_{\text{iter}} i} = 0 \\ & d_i \neq 1, \quad \text{if } \boldsymbol{\alpha}_{k_{\text{iter}} i} = \frac{1}{N_{k_{\text{iter}}}} \\ & \#\left\{ d_i | d_i \neq 0, 1 \leq i \leq N_{k_{\text{iter}}} \right\} = 2 \\ & d_i \in \{1, 0, -1\}, \quad 1 \leq i \leq N_{k_{\text{iter}}} \end{aligned} \tag{40}$$

where $\nabla f(\boldsymbol{\alpha}_{k_{\text{iter}}})$ denotes the gradient of the objective function for (39). The entries of $\boldsymbol{d}$ are shown to be -1, 1, and 0, and the nonzero elements in $\boldsymbol{d}$ indicate the index of selected working variables. To analytically solve variables in the working set, only two variables (i.e., pair variables) are selected into working set $\boldsymbol{\alpha}_{k_{\text{iter}} B}$. According to the constraints $\sum_i d_i = 0$ and $\sum_i y_{k_{\text{iter}} i} d_i = 0$ in (40), the two nonzero elements of $\boldsymbol{d}$ are $d_{i_1} = 1$ and $d_{i_2} = -1$, respectively, and the corresponding labels satisfy $y_{k_{\text{iter}} i_1} = y_{k_{\text{iter}} i_2}$. Therefore, the pair-index number of $\boldsymbol{d}$ with nonzero elements can be described as $S_{k_{\text{iter}}} = \{(i_1, i_2) | y_{k_{\text{iter}} i_1} = y_{k_{\text{iter}} i_2}, \boldsymbol{\alpha}_{k_{\text{iter}} i_1} < (1/N_{k_{\text{iter}}}), \boldsymbol{\alpha}_{k_{\text{iter}} i_2} > 0\}$, and the optimal pair variables in $S_{k_{\text{iter}}}$ that minimize the objective function of (40) are selected into the current working set, as shown in the following equation:

$$(i_1, i_2) = \arg\min_{(i_1, i_2)} \left\{ \nabla f\left(\boldsymbol{\alpha}_{k_{\text{iter}} i_1}\right) - \nabla f\left(\boldsymbol{\alpha}_{k_{\text{iter}} i_2}\right) | (i_1, i_2) \in S_{k_{\text{iter}}} \right\}. \tag{41}$$

After selecting working set $\boldsymbol{\alpha}_{k_{\text{iter}} B} = [\alpha_{k_{\text{iter}} i_1}, \alpha_{k_{\text{iter}} i_2}]$, it is necessary to calculate the step size d_{step} for updating variables for the working set. By implicitly embedding equality constraints into the objective function of (39), the resulting objective function is shown in the following equation:

$$\begin{aligned} & \min_{d_{\text{step}}} f\left(d_{\text{step}}\right) \\ & = \left(\boldsymbol{\alpha}_{k_{\text{iter}} B} + d_{\text{step}} \times [1, -1]\right)^T Q_{BB} \left(\boldsymbol{\alpha}_{k_{\text{iter}} B} + d_{\text{step}} \times [1, -1]\right) \\ & \quad + 2 \times \left(\boldsymbol{\alpha}_{k_{\text{iter}} B} + d_{\text{step}} \times [1, -1]\right)^T Q_{BN} \boldsymbol{\alpha}_N \end{aligned} \tag{42}$$

where $\boldsymbol{\alpha}_{k_{\text{iter}} B} + d_{\text{step}} \times [1, -1]$ denotes updated variables in the working set. To obtain the optimal d_{step} without considering the bound constraints of (39), let the derivatives of the objective function in (42) with respect to d_{step} be zero; then, we have

$$d^*_{\text{step}} = \frac{-[1, -1]^T [Q_{BB}, Q_{BN}] \left[\boldsymbol{\alpha}_{k_{\text{iter}} B}, \boldsymbol{\alpha}_N\right]}{[1, -1]^T Q_{BB} [1, -1]}. \tag{43}$$

d^*_{step} in (43) indicates the optimal solution of (42), while it could be outside the feasible region specified by the bound constraints in (39), i.e., the following conditions are not satisfied:

$$\begin{aligned} & 0 \leq \alpha_{k_{\text{iter}} i_1} + d_{\text{step}} \leq \frac{1}{N_{k_{\text{iter}}}} \\ & 0 \leq \alpha_{k_{\text{iter}} i_2} - d_{\text{step}} \leq \frac{1}{N_{k_{\text{iter}}}} \\ & \Rightarrow \\ & \quad - \min\left(\frac{1}{N_{k_{\text{iter}}}} - \alpha_{k_{\text{iter}} i_2}, \alpha_{k_{\text{iter}} i_1} \right) \leq d_{\text{step}} \\ & \quad \leq \min\left(\frac{1}{N_{k_{\text{iter}}}} - \alpha_{k_{\text{iter}} i_1}, \alpha_{k_{\text{iter}} i_2} \right). \end{aligned} \tag{44}$$

If d^*_{step} does not satisfy (44), the other d_{step} satisfied by (44) that minimizes the objective function $f(d_{\text{step}})$ of (42) will be selected as the final step-size. The value of $f(d_{\text{step}})$ can be calculated as follows:

$$f\left(d_{\text{step}}\right) = f\left(d^*_{\text{step}}\right) + \left(d_{\text{step}} - d^*_{\text{step}}\right)^2 [1, -1]^T Q_{BB} [1, -1]. \tag{45}$$

Since Q_{BB} belongs to a positive semidefinite matrix, we have $[1, -1]^T Q_{BB} [1, -1] \geq 0$, i.e., the larger the value of $(d_{\text{step}} - d^*_{\text{step}})^2$ is, the larger the value of $f(d_{\text{step}})$. Therefore, d_{step} bounded by (44) that is nearest to the optimal d^*_{step} will be selected as the final step size, as described in the following equation:

$$d_{\text{step}} = \begin{cases} -\min\left(\dfrac{1}{N_{k_{\text{iter}}}} - \alpha_{k_{\text{iter}} i_2}, \alpha_{k_{\text{iter}} i_1} \right), \\ \qquad \text{if } d^*_{\text{step}} < -\min\left(\dfrac{1}{N_{k_{\text{iter}}}} - \alpha_{k_{\text{iter}} i_2}, \alpha_{k_{\text{iter}} i_1} \right) \\ \min\left(\dfrac{1}{N_{k_{\text{iter}}}} - \alpha_{k_{\text{iter}} i_1}, \alpha_{k_{\text{iter}} i_2} \right), \\ \qquad \text{if } d^*_{\text{step}} > \min\left(\dfrac{1}{N_{k_{\text{iter}}}} - \alpha_{k_{\text{iter}} i_1}, \alpha_{k_{\text{iter}} i_2} \right) \\ d^*_{\text{step}}, \quad \text{otherwise.} \end{cases} \tag{46}$$

By applying the final d_{step}, $\boldsymbol{\alpha}_{k_{\text{iter}} B}$ is updated as follows:

$$\boldsymbol{\alpha}_{k_{\text{iter}} B} = \boldsymbol{\alpha}_{k_{\text{iter}} B} + d_{\text{step}} \times [1, -1]. \tag{47}$$

Perform (39)–(47) repeatedly until the following convergence criterion is met. Then, the current $\boldsymbol{\alpha}$ is output

$$\max_j \min_{(i_1, i_2)} \left\{ \nabla f\left(\alpha_{j i_1}\right) - \nabla f\left(\alpha_{j i_2}\right) | (i_1, i_2) \in S_j \right\} \leq \delta_d \tag{48}$$

where δ_d denotes a small positive number. The AHSTM-oriented decomposition algorithm has been finished. The implementation of the AHSTM-oriented decomposition algorithm is summarized in Algorithm 3.

V. Discussion of the Performance for AHSTM

Compared to existing classification algorithms that are used for fixed sources, the AHSTM is proposed to adapt to arbitrary combinations by only training once in Section III. In addition, to further emphasize the superiority of AHSTM, the classification performance of AHSTM for multisource object recognition is discussed elaborately. As mentioned above, the samples

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from case 1 (i.e., multisource samples) cannot always be available, and more obtained samples are from a single source (i.e., data from cases 2 and 3). For this situation, the existing methods can only utilize the samples from case 1 to train the model for multisource object recognition, discarding the available samples from cases 2 and 3. When multisource samples are not sufficient to train the classifier, the existing methods could overfit the insufficient samples from case 1, leading to a decrease in the classification performance. In comparison, the AHSTM can train a joint model using samples from case 1 and cases 2 and 3 and avoid overfitting insufficient samples from case 1. To further emphasize this advantage, a detailed analysis of the performance of AHSTM is given from the following two perspectives.

A. Performance Analysis of AHSTM From the Perspective of TL

Given training set $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, \mathcal{X}_2^{(im)}|_{m=1}^{M_2}, y_i\}|_{i=1}^{N_1}$ from case 1, $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, y_i\}|_{i=1}^{N_2}$ from case 2, and $\{\mathcal{X}_2^{(im)}|_{m=1}^{M_2}, y_i\}|_{i=1}^{N_3}$ from case 3, facing the situation in which multisource samples (i.e., N_1) are not sufficient to train the classifier, TL methods are adopted to reduce the impact of insufficient training samples on the performance of the classifier. TL [37]–[39] has been a research hotspot in the machine learning field in recent years and can apply the knowledge available in source domains to improve the performance of the model for a new yet related domain (i.e., target domain). The existing TL methods can be divided into two groups [37]: homogeneous TL, where data from the source domain and target domain lie in the same feature space, and heterogeneous TL, where data from the source domain and target domain lie in different feature spaces. In this article, the samples from case 1 (i.e., multisource samples) are represented as the target domain data, and the samples from cases 2 and 3 (i.e., single-source samples) are considered as the source domain data. It is obvious that the task that applies the single-source samples to improve the recognition performance of multisource samples can be considered to be a heterogeneous TL framework. Take two examples for heterogeneous TL. In [38], combined with weighted SVM, the samples in the source domain and target domain with different dimensions were mapped into a common space. These samples are reweighted iteratively to train a joint classifier to improve the classification performance of the target domain. In [39], to improve the classification performance for multiple tasks in different domains, the heterogeneous multitask metric learning framework is constructed to transform data from the different domains into a common subspace. Then, the classifiers are trained to identify the transformed data.

For the above heterogeneous TL methods, they always attempt to find related information between different domains by mapping samples from different domains into a common space. Then, they use the mapped samples to train the classifier to improve the classification performance of the target domain. For object recognition for arbitrary combinations, although the samples from different combinations present different dimensions, the heterogeneous tensors extracted from the same source are drawn from the same distribution. This possibility means that there is related information between different combinations. In this way, the proposed AHSTM utilizes samples from different combinations to jointly learn the parameters of shared projecting tensors $\{\mathcal{W}_{(r)}^{(m)}|1 \le r \le R, 1 \le m \le M_0\}$ to transfer the related information from cases 2 and 3 to improve the recognition performance for case 1. Considering that there still exists individuality for specific combinations, in addition to shared projecting tensors, the auxiliary projecting tensors $\mathcal{W}_{\text{aux}}^{(jm)}|_{j,m}$ are provided in AHSTM to further mine the useful information for each combination. Therefore, from the perspective of TL, the AHSTM can improve the recognition performance for multisource data by transferring the related information from single-source data.

B. Performance Analysis of AHSTM From the Perspective of a Feasible Region

Subsequently, we will analyze the performance advantage of AHSTM from the perspective of feasible regions for multisource object recognition.

For the training set from three cases, i.e., $\{\mathcal{X}_0^{(im)}|_{m=1}^{M_1+M_2}, y_i\}|_{i=1}^{N_1}$, $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, y_{2\ i}\}|_{i=1}^{N_2}$, and $\{\mathcal{X}_2^{(im)}|_{m=1}^{M_2}, y_{3\ i}\}|_{i=1}^{N_3}$, considering hard margin AHSTM with $\delta = 0$, i.e., removing ξ_{ji}, the corresponding optimization problem [see (28)] can be converted to the following equation:

$$\min_{\boldsymbol{w}_l^{(m)}|_{l=1,m=1}^{l_m,M_1+M_2},b,p} \frac{1}{2}\sum_{m=1}^{M_1+M_2}\sum_{r=1}^{R}\left\|\boldsymbol{w}_{(1\ r)}^{(m)}\circ,\ldots,\circ\boldsymbol{w}_{(l_m r)}^{(m)}\right\|_F^2 - \sum_{j=1}^{3}\nu_j p_j$$

$$\text{s.t. } y_{1\ i}\left(\sum_{m=1}^{M_0}\left(\sum_{r=1}^{R}\mathcal{X}_0^{(im)}\prod_{l=1}^{l_m}\times_l \boldsymbol{w}_{(lr)}^{(m)}\right)+b_1\right) \ge p_1 \quad 1\le i \le N_1 \tag{49a}$$

$$y_{2\ i}\left(\sum_{m=1}^{M_1}\left(\sum_{r=1}^{R}\mathcal{X}_1^{(im)}\prod_{l=1}^{l_m}\times_l \boldsymbol{w}_{(lr)}^{(m)}\right)+b_2\right) \ge p_2 \quad 1\le i \le N_2 \tag{49b}$$

$$y_{3\ i}\left(\sum_{m=M_1+1}^{M_1+M_2}\left(\sum_{r=1}^{R}\mathcal{X}_2^{(im)}\prod_{l=1}^{l_m}\times_l \boldsymbol{w}_{(lr)}^{(m)}\right)+b_3\right) \ge p_3 \quad 1\le i \le N_3. \tag{49c}$$

In the above hard margin AHSTM, δ is set to zero to simplify the analysis process. Suppose that the single-source samples $\{\mathcal{X}_1^{(im)}|_{m=1}^{M_1}, y_{2\ i}\}|_{i=1}^{N_1}$ and single-source samples $\{\mathcal{X}_2^{(im)}|_{m=1}^{M_2}, y_{3\ i}\}|_{i=1}^{N_1}$ are split from multisource samples $\{\mathcal{X}_0^{(im)}|_{m=1}^{M_1+M_2}, y_{1\ i}\}|_{i=1}^{N_1}$; then, we have

$$y_{2\ i}\left(\sum_{m=1}^{M_1}\sum_{r=1}^{R}\mathcal{X}_1^{(im)}\prod_{l=1}^{l_m}\times_l \boldsymbol{w}_{(lr)}^{(m)}+b_2\right) > p_2, \quad 1\le i\le N_1$$

$$y_{3\ i}\left(\sum_{m=M_1+1}^{M_1+M_2}\sum_{r=1}^{R}\mathcal{X}_2^{(im)}\prod_{l=1}^{l_m}\times_l \boldsymbol{w}_{(lr)}^{(m)}+b_3\right) \ge p_3, \quad 1\le i\le N_1$$

$$\Rightarrow y_{1\ i}\left(\sum_{m=1}^{M_1+M_2}\sum_{r=1}^{R}\mathcal{X}_0^{(im)}\prod_{l=1}^{l_m}\times_l \boldsymbol{w}_{(lr)}^{(m)}+b_2+b_3\right) \ge p_2+p_3. \tag{50}$$

This means that, if $\boldsymbol{w}_{(lr)}^{(m)}|_{l=1,r=1,m=1}^{l_m,R,M_1}$ and $\boldsymbol{w}_{(lr)}^{(m)}|_{l=1,r=1,m=M_1+1}^{l_m,R,M_1+M_2}$ can be used to classify samples from case 2 and case 3, which are split from the samples in case 1, respectively, then

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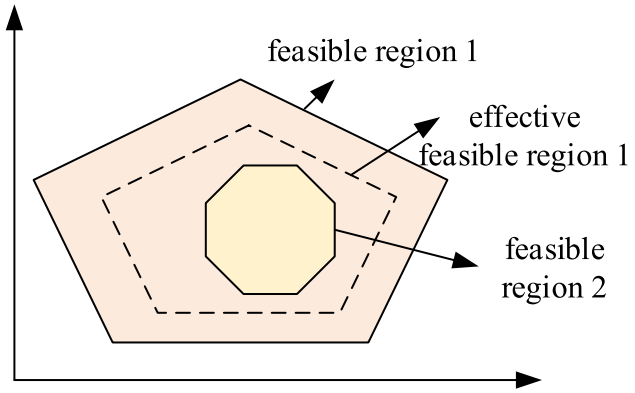


Fig. 3. Illustration of feasible region for projecting vectors defined by different constraints.

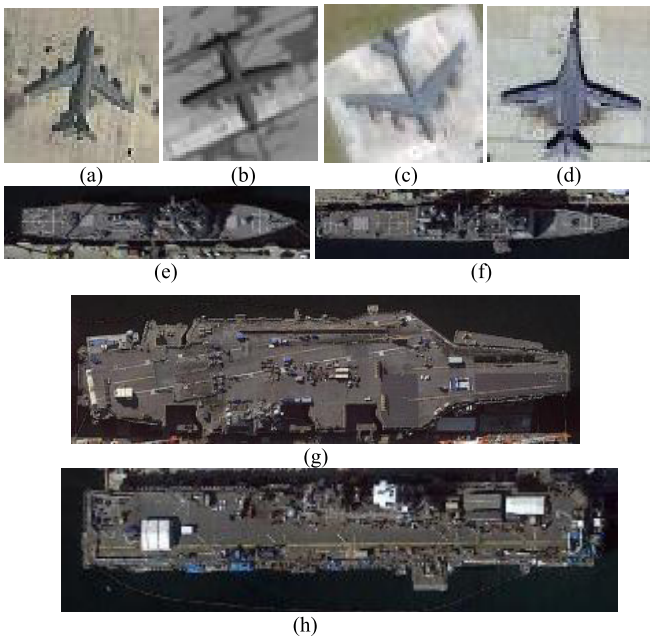


Fig. 4. Examples of slices in dataset 1 for objects with different types. (a)–(d) Airplane slices with four types. (e)–(h) Ship slices with four types.

$\boldsymbol{w}_{(lr)}^{(m)}|_{l=1,r=1,m=1}^{l_m,R,M_1+M_2}$ can be exploited to classify samples from case 1. This finding indicates that feasible region 2 of projecting vectors $\boldsymbol{w}_{(lr)}^{(m)}|_{l=1,r=1,m=1}^{l_m,R,M_1+M_2}$ defined by the constraints of (49b) and (49c) is the subset of feasible region 1 defined by the constraints of (49a), as illustrated in Fig. 3.

It is obvious that the projecting vectors $\boldsymbol{w}_{(lr)}^{(m)}|_{l=1,r=1,m=1}^{l_m,R,M_1+M_2}$ need to lie inside feasible region 2 for hard margin AHSTM. For soft margin AHSTM [i.e., the AHSTM described in (28)], since the slack variables ξ_{ji} exist, the projecting vectors $\boldsymbol{w}_{(lr)}^{(m)}|_{l=1,r=1,m=1}^{l_m,R,M_1+M_2}$ can lie outside the two feasible regions, but it could incur the extra cost generated by positive ξ_{ji}.

The existing methods can only utilize the samples from case 1 to achieve multisource object recognition but cannot use samples from cases 2 and 3. When samples from case 1 are not enough to train the classifier, feasible region 1 will become too large to characterize the effective solutions of the projecting vectors accurately. In comparison, the AHSTM can jointly utilize the samples from three cases to jointly learn the parameters of projecting vectors and impose the penalty on projecting vectors that deviate from feasible region 2. This approach can be roughly regarded as shrinking the size of feasible region 1, thereby reducing the occurrence of inadequate solutions, i.e., learning parameters from samples of three cases and avoiding overfitting the solution to insufficient multisource samples.

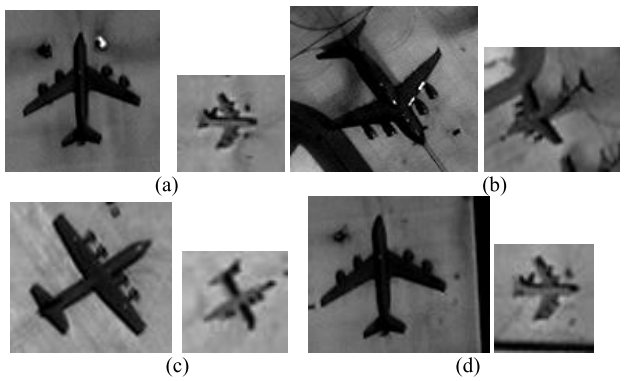


Fig. 5. Examples of multiangle and multiresolution airplane slices observed by Superview-1 and Jilin 1 satellites in dataset 2. (a)–(d) Paired multiangle multiresolution airplanes obtained from Superview-1 satellites (the left one) and Jilin 1 satellites (the right one).

TABLE II
DETAILED INFORMATION ON REMOTE SENSING IMAGES CONTAINING AIRPORT REGIONS

satellite	roll angle	resolution	acquired time
SuperView-1	-2.79 degree	0.5 meter	02.09.2020
SuperView-1	-13.62 degree	0.5 meter	07.30.2020
Jilin-1	33.11 degree	1 meter	10.07.2020
Jilin-1	-34.20 degree	1 meter	11.07.2020

VI. EXPERIMENTS AND ANALYSIS

Compared with typical SVM and STM methods, the experiments are conducted on three datasets that contain RSIs to verify the superiority of the proposed methods, and the detailed information of the three datasets is described as follows.

1) Dataset 1 consists of 100 airplane slices with four types and 107 ship slices with four types collected from public Google Earth service, whose spatial resolutions are equal to 1 m, of which the number of airplane slices of the four types is 17, 18, 10, and 51, respectively, and the number of ship slices of the four types is 51, 40, 8, and 8, respectively. Examples of slices with different types from dataset 1 are displayed in Fig. 4.
2) Dataset 2 comprises multiresolution and multiangle airplane slices of three types obtained by Superview-1 commercial satellites with roll satellite angles less than 20° and Jilin 1 commercial satellites with roll satellite angles exceeding 20°. These airplane slices are extracted from four RSIs of the airport region, and detailed information on the four RSIs is given in Table II.
   Since these RSIs are acquired in different time phases, some airplanes in the airport are observed by two

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satellites, and some airplanes are observed by only one satellite, resulting in three combinations of situations: 1) case 1: the acquired data are observed by two satellites (multisource samples); 2) case 2: the acquired data are observed by the Superview-1 satellite (single-source samples); and 3) case 3: the acquired data are observed by the Jilin 1 satellite (single-source samples). The numbers of samples for the three cases are equal to 28, 32, and 37. Examples of extracted slices for different objects in dataset 2 are listed in Fig. 5.

3) Dataset 3 consists of 31 paired optical-SAR (i.e., SAR) ship slices of four types and 107 single-source optical ship slices of four types collected from Google Earth service and Terra SAR satellite data, and the spatial resolutions of optical slices and SAR slices are both 1 m. Thus, there are two combinations for dataset 3, i.e., case 1: the obtained data are paired optical-SAR samples; case 2: the obtained data are optical samples. Examples of paired optical-SAR ship slices in dataset 3 are displayed in Fig. 6.

Dataset 1 is used to evaluate the performance of HSTM. Since dataset 2 and dataset 3 contain paired multisource slices (i.e., multiangle multiresolution slices and optical-SAR slices) and single-source slices, they are exploited to examine the effect of HSTM and AHSTM, especially for AHSTM under arbitrary combination situations.

The experiments are conducted in five parts. The construction of used heterogeneous feature tensors is given in Section VI-A, and the parameter setting of HSTM is analyzed in Section VI-B. In Section VI-C, the recognition performance of HSTM is evaluated in terms of a specific combination situation compared with typical SVM and STM methods by using dataset 1. In Section VI-D, the recognition performance of AHSTM is evaluated in terms of arbitrary combination situations compared with the proposed HSTM and typical SVM and STM methods by using dataset 2 and dataset 3. In Section VI-E, the superiority of the AHSTM is evaluated compared with the deep learning methods for different combinations of dataset 2 and dataset 3. In addition, to demonstrate the advantage of the proposed decomposition-based AHSTM training method, time consumption comparison experiments between the proposed decomposition method and typical optimization algorithms, i.e., the interior-point method and the active-set method, are conducted on dataset 1 and dataset 2 in Section VI-E.

All the simulations are implemented running on an i5-7300HQ processor and 8-GB memory with a Windows 10 operating system.

A. Construction of Heterogeneous Feature Tensors

To effectively extract the rich information contained in multisource remote sensing data, multiple feature tensors are constructed to extract different attributes of objects used for object recognition, as shown in the following.

1) Gabor Features Tensor: Using different Gabor filters [see (51)], Gabor features [40] are used to describe the texture characteristics of objects with multiple scales and orientations

$$G_{s,d}(x,y) = G_{\overrightarrow{\kappa}}\left(\overrightarrow{x}\right) = \frac{\left\|\overrightarrow{\kappa}\right\|}{\sigma^2}\cdot e^{-\frac{\|\overrightarrow{\kappa}\|^2\cdot\|\overrightarrow{x}\|^2}{2\sigma^2}}\cdot e^{i\overrightarrow{\kappa}\cdot\overrightarrow{x}} - e^{-\frac{\sigma^2}{2}} \quad (51)$$

where $\overrightarrow{x} = (x, y)$ and $\overrightarrow{\kappa} = (\pi/2 \times 2^s) \cdot [\cos((\pi d/8)), \sin((\pi d/8))]$ denote the spatial domain coordinate and the frequency vector, respectively. In the experiments, the scale parameter s and the orientation parameter d are set to $\{0, 1, 2, 3, 4\}$ and $\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$, respectively. Thus, the obtained 40 Gabor filters are used to extract Gabor features and store the responses of the Gabor filters to the third-order tensor $F_{\text{gabor}} \in \mathbb{R}^{w\times h\times 40}$, where w and h denote the width of the image and the height of the image, respectively.

2) Local Binary Pattern Features Tensor: Local binary pattern (LBP) features [41] record the differences between the values of the current pixel and its neighboring pixels. In experiments, by setting the radius of the neighbors as $\{1, 2, 3\}$, the LBP features with different scales are extracted and stored as a tensor $F_{LBP} \in \mathbb{R}^{w\times h\times 3}$.

3) Histograms of the Oriented Gradient (HoG) Features Tensor: HoG features [42] are the representative contour features, which have been widely applied in object detection and

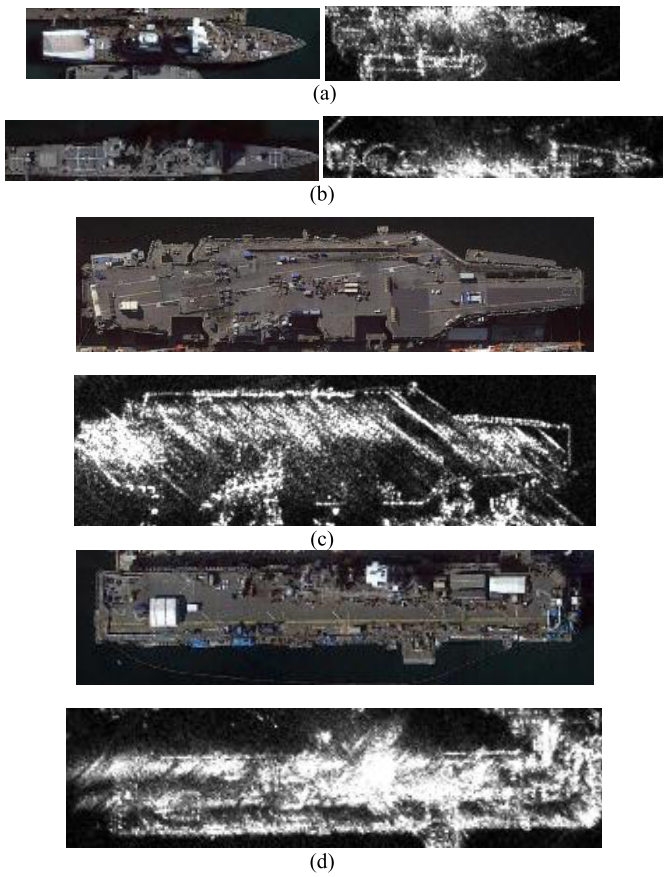


Fig. 6. Paired optical-SAR object slices for ships with different types. (a) Paired optical-SAR object slices for the ship with type 1. (b) Paired optical-SAR object slices for the ship with type 2. (c) Paired optical-SAR object slices for the ship with type 3. (d) Paired optical-SAR object slices for the ship with type 4.

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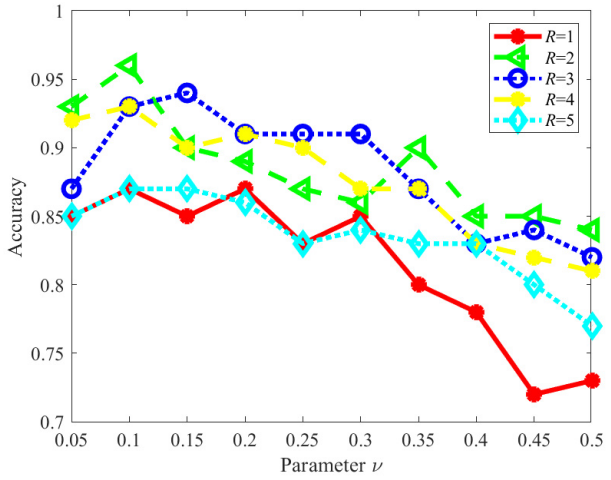


Fig. 7. Recognition accuracy of HSTM under different parameter settings.

recognition tasks. In the experiments, the size of the block and the number of bins in the histogram for the HoG features are set to 2×2 and 9, respectively. Thus, by adjusting the size of the cell blocks in the HoG features to $\{4 \times 4, 8 \times 8, 16 \times 16\}$, HoG features with different cell sizes are extracted and stored as feature tensors $F^1_{HoG} \in \mathbb{R}^{w_1 \times h_1 \times 9}$, $F^2_{HoG} \in \mathbb{R}^{w_2 \times h_2 \times 9}$, and $F^3_{HoG} \in \mathbb{R}^{w_3 \times h_3 \times 9}$, where $[w_1, w_2, w_3]$ and $[h_1, h_2, h_3]$ denote the number of cells in the horizontal direction and the number of cells in the vertical direction for the image slice, respectively.

4) Morphology Features Tensor: The top-hat transform and bottom-hat transform for morphology processing are used to extract the partial details of the information in the image [43]. By applying the two operations, the extracted morphology features are stored as feature tensors $F^1_{\text{mor}} \in \mathbb{R}^{w \times h}$ and $F^2_{\text{mor}} \in \mathbb{R}^{w \times h}$.

5) Scattering Features Tensor: The scattering characteristic of SAR images reflects the structure and material information of objects. In the experiments, the gray-level cooccurrence matrixes (GLCMs) [43] with four scanning types for pair pixels ($\{[2, 0], [0, 2], [1, 1], [1, -1]\}$) are utilized to extract the scattering feature tensor $F_{\text{GLCM}} \in \mathbb{R}^{8 \times 8 \times 4}$, where the dimensions of the first order and second order denote the quantized gray level of the GLCM features.

To obtain effective recognition results, for airplane recognition using optical images, all the feature tensors are exploited except for the scattering feature tensor. For ship recognition using optical images, HoG feature tensors and morphology feature tensors are utilized as inputs. For ship recognition using SAR images, only scattering feature tensors are used in the experiments.

B. Analysis of Parameter Setting for HSTM and AHSTM

The main parameters of HSTM are ν and R, which are used to control the number of support tensors and limit the maximum CP rank of the projecting tensor, respectively. Since AHSTM can be considered a generalized case for HSTM, for convenience, the parameter settings for ν and R can only be analyzed using HSTM. In this way, by applying 100 airplane slices with four types in dataset 1, HSTMs with different ν and R are used to obtain the corresponding recognition results, as shown in Fig. 7.

It is observed from Fig. 7 that HSTM obtains different recognition accuracies under different parameters and achieves the best results when $\nu = 0.1$ and $R = 2$. Since the parameter ν determines the upper bound of errors, a large ν will increase the empirical error and weaken the recognition results, and it is seen that the recognition accuracy decreases significantly under $\nu > 0.3$. In addition, it can be seen that as R increases, the recognition accuracy is first improved and then decreased. The reason is that the projecting tensor with rank 1, i.e., $R = 1$, cannot effectively describe the accuracy classification hyperplane, and the projecting tensor with a large rank easily overfits the training set.

In addition to the parameters ν and R, the AHSTM includes the parameter δ. To demonstrate the impact of δ on the accuracy of AHSTM, the classification results of three cases

TABLE III
CLASSIFICATION ACCURACY OF AHSTM WITH DIFFERENT PARAMETER δ FOR DATASET 2

δ	accuracy (case 1)	accuracy (case 2)	accuracy (case 3)
0	**96.43**	**90.63**	67.57
0.1	92.86	**90.63**	67.57
0.2	89.29	87.50	**70.27**
0.3	89.29	87.50	67.57

TABLE IV
RECOGNITION ACCURACIES OF AIRPLANE SLICES FOR DIFFERENT METHODS UNDER N-FOLD CROSS-VALIDATION

	methods	R	C	ν	accuracy(%)
N=5	HSTM	3	-	0.1	**90**
	C-SVM	-	2^3	-	89
	$\nu-$SVM	-	-	0.1	**90**
	C-STM	-	2^0	-	78
N=10	HSTM	2	-	0.1	**96**
	C-SVM	-	2^3	-	90
	$\nu-$SVM	-	-	0.1	89
	C-STM	-	2^4	-	82

TABLE V
RECOGNITION ACCURACIES OF SHIP SLICES FOR DIFFERENT METHODS UNDER N-FOLD CROSS-VALIDATION

	methods	R	C	ν	accuracy(%)
N=5	HSTM	2	-	0.1	**85.05**
	C-SVM	-	2^2	-	79.44
	$\nu-$SVM	-	-	0.2	79.44
	C-STM	-	2^0	-	70.1
N=10	HSTM	3	-	0.2	**87.85**
	C-SVM	-	2^3	-	85.05
	$\nu-$SVM	-	-	0.1	85.05
	C-STM	-	2^4	-	82

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for dataset 2 are obtained using AHSTM with different δ's, as shown in Table III.

Note that dataset 2 contains insufficient samples for each case, where there are 28, 32, and 37 samples for case 1, case 2, and case 3, respectively. Therefore, for the situation where lacking sufficient samples, it is proper to use a small δ so that the parameters of the classification hyperplane can be learned adequately from the samples from different combinations. In this way, it can be found that AHSTM with $\delta = 0$ obtains the best results for case 1 and case 2, and AHSTM with $\delta = 0.2$ obtains the best results for case 3. Although better detection results are obtained when the parameter delta is 0, this only indicates that using only the shared projection tensor is enough in a given case. To obtain the best results, an effective strategy is to set reasonable parameter δ for different cases in order to obtain optimal results. In addition, note that, when $\delta = 0.3$, the corresponding accuracy is smaller than using a smaller δ because a larger δ may lead to a decrease in weight for shared projecting tensors and enable the classifier to overfit the training set of a specific case. According to the parameter analysis experiments, better recognition results may be obtained for the proposed HSTM and AHSTM when $\nu < 0.3$, $R > 1$, and $\delta \leq 0.2$.

C. Evaluation of HSTM Performance

Experiments were conducted on 100 airplane slices and 107 ship slices in dataset 1 to compare HSTM with typical SVM and STM. To fully use dataset 1 to accurately evaluate the performance of different methods, N-fold cross-validation ($N \in \{5, 10\}$) was adopted to divide the dataset into N subsets, and then, N-1 subsets were selected alternatively for training and the remaining subset for testing. Since dataset 1 contains multiclass objects, the one-against-one strategy [10] was introduced to ensure that HSTM, SVM, and STM can process the multiple classification problem. In addition, due to the input requirements of the C-SVM and ν-SVM methods, the vectorization operation was applied to convert the heterogeneous feature tensors to vectors. For C-STM, since it requires the input data to have a single-tensor structure, the extracted heterogeneous feature was expanded and filled with zeros elements to multiple third-order tensors with the same sizes, and then, those third-order tensors are connected along the new dimension to construct a fourth-order feature tensor as the input data of the STM. Moreover, to ensure a fair comparison, all the methods selected the optimal parameters from $C \in \{2^0, 2^1, 2^2, 2^3, 2^4\}$, $\nu \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, and $R \in \{1, 2, 3\}$ to obtain the best detection results. The accuracies of different methods for airplane recognition and ship recognition are displayed in Tables IV and V, respectively.

Note that a large N indicates more training samples, and it can be found that a larger accuracy is obtained under a larger N for the same method. In addition, the recognition accuracy of the airplane slices is larger than that of the ship slices since ship slices contain more interference than airplane slices. Because the conversion from heterogeneous tensors to a single tensor drastically damages the structure of the input, STM achieves the worst recognition accuracy. Compared with SVM and STM, which rely on the preprocessing method to convert input data to vectors and single tensors, HSTM can identify the samples in heterogeneous tensor space, avoiding the loss of structural information for input data. Therefore, HSTM outperforms the other methods in terms of recognition accuracy.

TABLE VI

RECOGNITION ACCURACIES OF DIFFERENT METHODS FOR EACH CASE USING DATASET 2

methods	parameters setting	accuracy (case 1)	accuracy (case 2)	accuracy (case 3)
AHSTM $\delta = 0$	R=3 $\boldsymbol{\nu} = [0.1, 0.1, 0.1]$	**96.43**	90.63	**67.57**
AHSTM $\delta = 0.1$	R=3 $\boldsymbol{\nu} = [0.1, 0.1, 0.1]$	92.86	90.63	**67.57**
HSTM	R=2 $\nu = 0.1$	92.86	-	-
HSTM	R=2 $\nu = 0.1$	-	**93.75**	-
HSTM	R=2 $\nu = 0.1$	-	-	64.86
C-SVM	$C = 2^3$	82.14	-	-
C-SVM	$C = 2^2$	-	81.25	-
C-SVM	$C = 2^3$	-	-	62.16
ν−SVM	$\nu = 0.1$	82.14	-	-
ν−SVM	$\nu = 0.2$	-	84.38	-
ν−SVM	$\nu = 0.1$	-	-	62.16
C-STM	$C = 2^2$	89.29	-	-
C-STM	$C = 2^1$	-	84.38	-
C-STM	$C = 2^2$			**67.57**

TABLE VII

RECOGNITION ACCURACY OF DIFFERENT METHODS FOR DIFFERENT COMBINATIONS OF SITUATIONS USING DATASET 3

methods	parameters setting	accuracy (case 1)	accuracy (case 2)
AHSTM $\delta = 0.1$	R=2 $\boldsymbol{\nu} = [0.2, 0.2, 0.2]$	90.32	**82.15**
AHSTM $\delta = 0$	R=2 $\boldsymbol{\nu} = [0.1, 0.1, 0.1]$	**96.77**	79.44
HSTM	R=1 $\nu = 0.2$	87.10	-
HSTM	R=1 $\nu = 0.2$	-	78.50
C-SVM	$C = 2^3$	74.19	-
C-SVM	$C = 2^2$	-	72.90
ν−SVM	$\nu = 0.1$	77.42	-
ν−SVM	$\nu = 0.1$	-	78.79
C-STM	$C = 2^2$	74.19	-
C-STM	$C = 2^0$	-	66.36

D. Evaluation of the AHSTM Performance With Typical SVM and STM Methods

To demonstrate the superiority of AHSTM for object recognition under arbitrary combination situations compared with C-SVM, ν-SVM, C-STM, and HSTM, comparison experiments are conducted on dataset 2 and dataset 3.

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1) Evaluation of the Accuracy of AHSTM With Typical SVM and STM Methods: In the Experiments, the N-fold cross-validation method is adopted to split samples from three cases into training sets and test sets to examine the accuracies for different recognition methods under three cases. Since HSTM, C-SVM, ν-SVM, and C-STM cannot adapt to arbitrary combination situations, these methods must construct multiple sets of model parameters (using the same training set generated from N-fold cross-validation) to address different cases. In addition, according to the input requirements of the C-SVM, ν-SVM, and C-STM, the same preprocessing operators are adopted to convert the extracted heterogeneous feature tensors to vectors or single tensors, as in Section VI-C. Thus, all the methods select the optimal parameters from $C \in \{2^0, 2^1, 2^2, 2^3, 2^4\}$, $\nu \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, and $R \in \{1, 2, 3\}$ to obtain the best recognition results, as shown in Table VI.

Note that the spatial resolution of images from the Superview-1 satellite (0.5 m) is larger than that from the Jilin 1 satellite (1 m), and it is easy to obtain better recognition results for Superview-1 satellite data. Therefore, it can be observed that the recognition results for case 2 are better than those for case 3. In addition, it is observed from the above table that AHSTM with $\delta = 0$ and $\delta = 0.1$ obtains effective recognition results for both cases because AHSTM can adapt to arbitrary combination situations effectively by using a single model. Moreover, AHSTM obtains better recognition results for case 1 (i.e., multisource object recognition) than HSTM because AHSTM can improve the multisource recognition results by exploiting the related information of different combinations compared with HSTM, which only uses samples from case 1. In addition, AHSTM obtains the best results among all the methods. For the comparison methods, they achieve worse recognition results than HSTM since the preprocessing for C-SVM, ν-SVM, and C-STM will lose the structural information of the input data.

Subsequently, compared with C-SVM, ν-SVM, C-STM, and HSTM, AHSTM is further verified using the multisource ship slices in dataset 3.

Dataset 3 contains paired optical-SAR ship slices and single optical ship slices. Using the same preprocessing method and parameter selection method as the experiments of dataset 2, the obtained recognition results for different methods under twofold cross-validation are displayed in Table VII.

It is observed from the above table that AHSTM can adapt to arbitrary combination situations using a single model. Avoiding the loss of structural information caused by data conversion used in other comparison methods, HSTM and AHSTM can separate samples in heterogeneous tensor space and achieve better recognition results than other competitors. It is found that AHSTM obtains a significant improvement in the recognition accuracy of case 1 compared to HSTM. This finding verifies that AHSTM can capture useful information between different combinations to improve the recognition results compared with HSTM. In addition, although it is found that AHSTM without auxiliary projecting tensors (i.e., $\delta = 0$) obtains the best results for case 1, it is also found that AHSTM with auxiliary projecting tensors ($\delta = 0.1$) obtains the best results among all of the comparison methods for case 2.

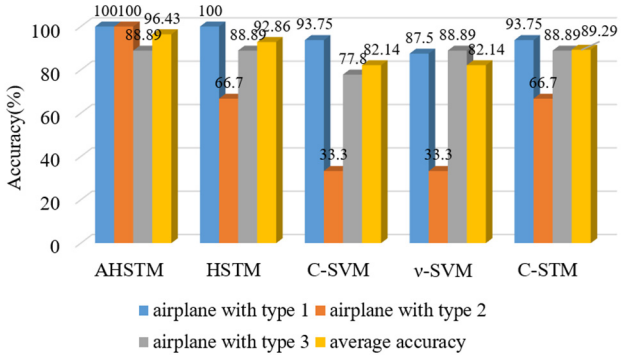


Fig. 8. Recognition accuracies of each type of airplanes under case 1 for different methods under twofold cross-validation.

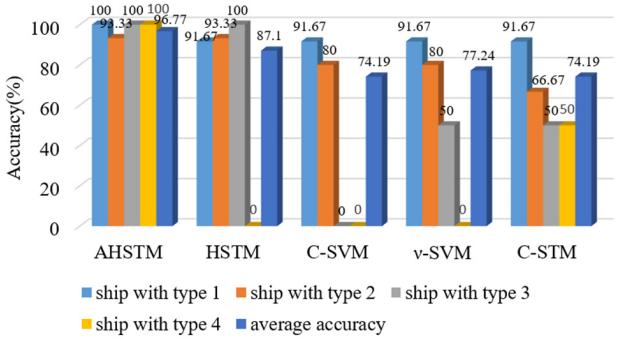


Fig. 9. Recognition accuracies of each type of ships under case 1 for different methods under N-fold cross-validation.

Therefore, it can be concluded that the auxiliary projecting tensors with positive δ are helpful to improve the recognition results for some cases. The above experiments indicate that AHSTM can adapt to arbitrary combination situations and achieve effective recognition results for multisource data.

2) Analyze the Impact of Different Classes of Objects on the Performance of AHSTM: To further analyze the performance of AHSTM for different classes of airplanes, using the same optimal parameters as in Table VI, the accuracies of the proposed AHSTM and HSTM and the other competitors for each class of airplanes using dataset 2 under case 1 (i.e., Object recognition using multisource data) are further counted in Fig. 8.

Compared with HSTM that can only use the multisource samples (i.e., samples from combination 1) to learn the parameters of classification hyperplane, the AHSTM can utilize samples jointly from different combinations to improve the effectiveness of the recognition results. In this way, it can be found that the accuracy of AHSTM for each class is greater than that of the HSTM. Since there are only three paired multisource samples for airplanes with type 2, it is difficult to train the parameters of comparison methods adequately. Nevertheless, AHSTM can train the classification hyperplane jointly with the multisource samples and single-source samples, preventing the model from overfitting to the small number of multisource samples. Therefore, it is noted that there is a significant improvement of AHSTM in the

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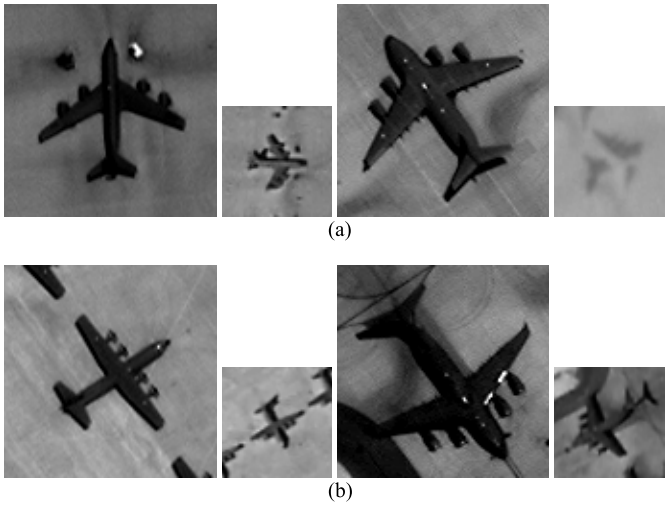


Fig. 10. Paired multiangle and multiresolution airplane slices containing different backgrounds. (a) Multiangle and multiresolution airplane slices containing smooth background. (b) Multiangle and multiresolution airplane slices containing the boundary of the airport or other object interferences.

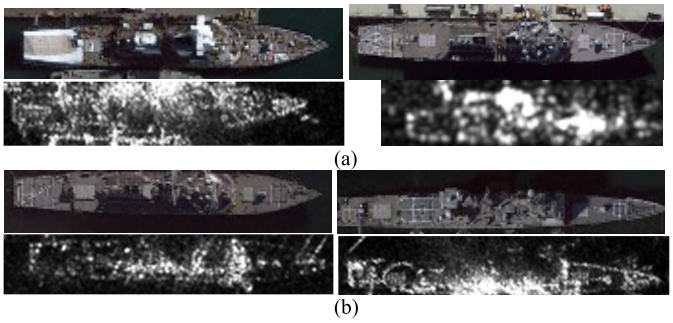


Fig. 11. Paired optical and SAR slices containing different backgrounds. (a) Paired optical and SAR slices containing dock. (b) Paired optical and SAR slices without a dock in the background.

recognition accuracy for airplanes with type 2. Moreover, from Fig. 8, it is seen that the AHSTM obtains the best accuracy for each class of objects among all the methods. It indicates that the effect of AHSTM for recognizing objects with different categories.

Besides, we compare the accuracies of AHSTM for each class of ships with the other methods under case 1 using dataset 3, as shown in Fig. 9.

Note that dataset 3 contains samples from two cases, where case 1 and case 2 denote the paired optical and SAR samples and single SAR samples, respectively. From Fig. 9, it can be found that the AHSTM obtains better results than the other methods for each class of ships, especially for ships with type 3 and ships with type 4. The reason is that there are only four paired optically and SAR samples for ships with type 3 and ships with type 4. For HSTM and the other classification methods, it can only use the samples from case 1 to learn the parameters of the hyperplane, ignoring the samples from case 2. In comparison, the AHSTM can learn the parameters of hyperplane using samples from two cases jointly. This finding implies that the shared projecting tensor of AHSTM

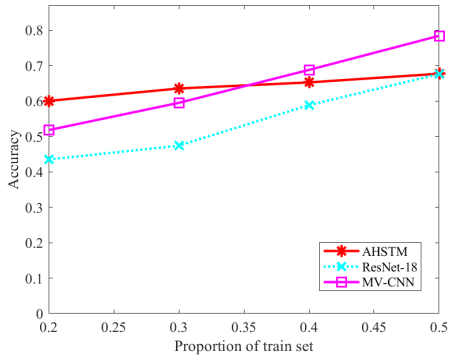


Fig. 12. Recognition accuracies of AHSTM and deep learning methods for different training set sizes under case 3 of dataset 2.

can extract useful information from the samples of different cases and, thus, improve the classification results.

3) Analyze the Impact of Different Backgrounds on the Performance of AHSTM: To further investigate whether the background of the slices affects the performance of AHSTM, different object slices were grouped according to the contained background to record the accuracy separately to analyze the impact of background differences on the performance of AHSTM.

For multiangle and multiresolution airplane slices, they can be divided into two groups, where the first group consists of the airplanes slices that containing the relatively smooth background [see Fig. 10(a)], and the second group consists of the airplanes slices that containing apron boundaries or other objects interference [see Fig. 10(b)]. Then, the performance of AHSTM is evaluated to obtain the recognition accuracy for these two groups of multiangle and multiresolution airplane slices compared with the other classification methods. The related parameters of different classification methods are the same as the parameters used in Table VI, and the obtained recognition accuracies for airplanes with different backgrounds are displayed in Table VIII.

Here, P_a denotes the average accuracy. It is seen from Table VIII that different methods obtain effective results for airplanes slices with smooth backgrounds because there are unobvious interferences in these slices. In contrast, different methods get lower accuracy for airplanes slices with object interferences. It is remarkable that the AHSTM can improve the accuracy for airplanes slices with object interferences compared with HSTM because the shared projecting tensors and auxiliary projecting tensors can mine the useful information of different combinations of multisource heterogeneous data. In conclusion, the AHSTM can obtain the best results for multiangle and multiresolution airplanes slices containing different backgrounds among all the recognition methods. It further demonstrates the effect of the AHSTM for multisource object recognition.

In addition, the experiments are further conducted on paired optical and SAR ship slices (i.e., samples of case 1 for dataset 3) containing different backgrounds to verify the effect of the AHSTM. These paired optical and SAR ship slices can

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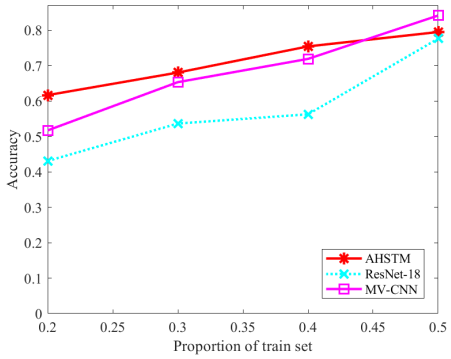


Fig. 13. Recognition accuracies of AHSTM and deep learning methods for different training set sizes under case 2 of dataset 3.

TABLE VIII

RECOGNITION ACCURACIES OF AIRPLANE SLICES WITH DIFFERENT BACKGROUNDS UNDER CASE 1 OF DATASET 2 FOR DIFFERENT METHODS USING TWO-FOLD CROSS-VALIDATION

	AHSTM	HSTM	*C*-SVM	$\nu-$SVM	*C*-STM
airplanes slices with smooth backgrounds	**100**	**100**	89.47	89.47	94.74
airplanes slices with object interferences	**88.89**	77.78	66.67	66.67	77.78
P_a(%)	**96.43**	92.86	82.14	82.14	89.29

TABLE IX

RECOGNITION ACCURACIES OF SHIP SLICES WITH DIFFERENT BACKGROUNDS UNDER CASE 1 OF DATASET 3 FOR DIFFERENT METHODS USING TWO-FOLD CROSS-VALIDATION

	AHSTM	HSTM	*C*-SVM	$\nu-$SVM	*C*-STM
ship slices without dock	**100**	**100**	80	90	70
ship slices containing dock	**95.45**	81.82	63.64	63.64	68.18
P_a(%)	**96.77**	87.10	74.19	77.42	74.19

be divided into two groups, where the first group consists of paired optical and SAR ship slices containing dock [see Fig. 11(a)], and the second group consists of paired optical and SAR ship slices without dock [see Fig. 11(b)]. Using *N*-fold cross-validation and the same parameter setting as in Table VII, the accuracies of different methods for paired optical and SAR ship slices containing different backgrounds are displayed in Table IX.

It is seen from Table IX that the accuracy of ship slices without dock is higher than that of ship slices containing dock for different recognition methods because the dock as the interference in the background may affect the recognition results. Note that AHSTM obtains better recognition results for ship slices with different backgrounds than the other methods. This finding indicates that the AHSTM is effective for multisource ship recognition using different backgrounds of slices.

TABLE X

RECOGNITION ACCURACY OF AHSTM AND DEEP LEARNING METHODS FOR DIFFERENT COMBINATIONS USING DATASET 2

	accuracy (case 1)	accuracy (case 2)	accuracy (case 3)
AHSTM	**96.43**	**90.63**	67.57
ResNet-18	89.29	75.00	67.57
MV-CNN	92.86	87.50	**78.38**

TABLE XI

RECOGNITION ACCURACY OF AHSTM AND DEEP LEARNING METHODS FOR DIFFERENT COMBINATIONS USING DATASET 3

	accuracy (case 1)	accuracy (case 2)
AHSTM	**96.77**	79.44
ResNet-18	74.19	77.57
MV-CNN	83.87	**84.11**

E. Evaluation of the AHSTM With Typical Deep Learning Methods

Considering that the deep learning methods have achieved significant success for various object recognition tasks, it is necessary to evaluate the performance of the AHSTM compared with the typical deep learning methods using dataset 2 and dataset 3, including deep residual networks (i.e., ResNet-18) [44] and multiview CNNs (MV-CNNs) [45].

In detail, since the ResNet-18 only accepts a single image as input, the paired multisource slices are concatenated into larger size slices to meet its input requirements. For MV-CNN, it can combine information from the input of multiple slices to predict the class of the object. To facilitate a fair comparison, both the AHSTM and deep learning methods adopt the twofold cross-validation strategy. To obtain a reliable recognition for the deep learning methods, the data augmentation method (i.e., scale transform and rotation transform [46]) is adopted to extend the training samples by three times. Then, the batch size and the learning rate are set to 2 and 5×10^{-4}, respectively. The resulted recognition accuracy for multiangle and multiresolution airplane recognition and optical and SAR ship recognition under different cases is displayed in Tables X and XI, respectively.

From the above tables, it can be found that the AHSTM obtains better results for multisource airplane and ship recognition (i.e., the accuracy of case 1) than the deep learning methods. Because there are a small number of samples for case 1, the deep learning methods are easy to overfit to the insufficient multisource samples. In addition, it is also seen that the accuracy of deep learning under case 2 is higher than that under case 1 for dataset 3 (see Table XI). The reason is that there are more samples for case 2 (i.e., 107) compared with case 1 (i.e., 31) to support the training of deep learning methods. Moreover, it is seen that the accuracy of MV-CNN is higher than that of ResNet-18 for different cases. Furthermore, it is observed that the MV-CNN obtains the best results for airplanes recognition using Jilin-1 data (i.e., case 3 of dataset 2) and ships recognition using optical data (i.e., case 2 of dataset 3). The reason is that both these two cases have a

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larger number of samples than other cases. To demonstrate the superiority of the AHSTM for a small-sample situation, the recognition accuracies of AHSTM and deep learning methods are further compared for different training set sizes under case 3 of dataset 2 and case 2 of dataset 3, respectively, as shown in Figs. 12 and 13.

The X-axis indicates the proportion of training samples occupying the total number of samples. From the above figures, it can be found that our AHSTM can obtain better accuracy for case 3 of dataset 2 when the proportion of the training set is less than 0.3, and AHSTM can obtain better accuracy for case 2 of dataset 3 when the proportion of training set is less than 0.4. This finding indicates the advantage of AHSTM for object recognition under a small-sample situation. More importantly, it is worth noting that these deep learning methods cannot adapt to arbitrary combinations of multisource data and have to train multiple networks separately for each case. In contrast, the AHSTM can handle different combinations of multisource data by only training once. It is concluded that AHSTM is able to adapt to arbitrary combinations of multisource data and obtains better recognition results compared to deep learning methods when lacking sufficient samples.

F. Comparison of the AHSTM-Oriented Decomposition Algorithm and Typical Optimization Solving Algorithms in Terms of Time Consumption

In this section, the time consumption of the proposed decomposition method is evaluated for solving the dual problem of HSTM and AHSTM compared with the typical optimization solving method (i.e., the interior point method [35] and the active-set method [36]) using dataset 1 and dataset 2.

In detail, dataset 1 and dataset 2 are utilized to examine the time consumption of solving the dual problem for HSTM and AHSTM, respectively. The parameter δ_d used in the convergence criterion of the proposed decomposition method is set to 10^{-6}, and the termination tolerance on the first-order optimality for the interior-point method and the active-set method is set to 10^{-6}. Applying different sizes of training sets, the corresponding time consumption of the different optimization methods for solving the dual problem of HSTM and AHSTM is displayed in Figs. 14 and 15, respectively.

In Figs. 14 and 15, the X-axis indicates the proportion of the training set to the total dataset. It is observed that the proposed decomposition method reduces the time consumption by more than 75% for both HSTM and AHSTM compared with the interior point method and the active set method under different sizes of training sets because the proposed decomposition method can utilize analytic small QP problems sequentially instead of the numerical large optimization problem used in typical optimization methods. In addition, since the size of matrix Q [see (28)] increases quadratically as the size of the training set increases, it is observed that the proposed decomposition method consumes a much shorter performance time compared with the comparison methods when the size of the training set becomes larger. This finding indicates that the proposed decomposition method is more effective in training HSTM and AHSTM compared with typical optimization solving algorithms.

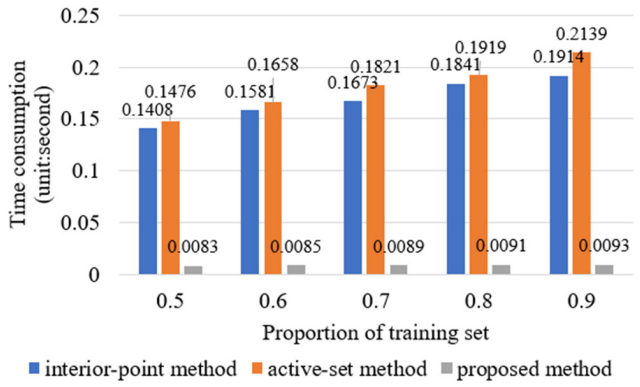


Fig. 14. Average time consumption of solving the dual problem once for HSTM under different methods using airplanes of dataset 1.

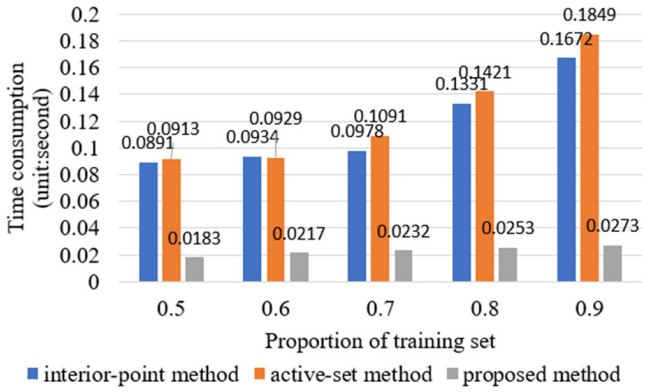


Fig. 15. Average time consumption of solving the dual problem once for AHSTM under different methods using dataset 2.

VII. Conclusion

To address arbitrary combinations of multisource heterogeneous remote sensing data describing objects and obtain excellent recognition results, three main contributions are made in this article, as summarized in the following.

1) Compared to the SVM and STM methods, which require the input to have a vector or single-tensor structures, the proposed HSTM accepts data with more structures as input, including vector, single-tensor, and heterogeneous tensor data. When processing multisource remote sensing data, it utilizes multitensor rank-R projecting mapping to separate samples in heterogeneous tensor space, avoiding the loss of structure information caused by data conversion of traditional SVM and STM. In addition, compared with the higher rank projecting tensor in [22], [28], and [29], the rank-R projecting in HSTM can limit the rank of the projecting tensor and eliminate the calculation of the inverse matrix. In addition, the HSTM can be considered to be a natural extension of ν-SVM and ν-STM. When the input is

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represented as a vector or single tensor, the HSTM degenerates to ν-SVM and ν-STM, respectively.

2) To recognize objects through observation data from arbitrary combinations of satellites with multiple types, the existing methods must train the model separately for each combination. In comparison, the proposed AHSTM can adapt to arbitrary combination situations by training only once. More importantly, AHSTM can improve the recognition accuracy compared with HSTM for multisource data by mining the related information intercombinations. In this way, AHSTM is not only an extension of existing STM methods but also an important improvement from the existing specific source-based classification algorithms to arbitrary combinations of multisource data-based classification algorithms.
3) When using existing optimization algorithms (e.g., the interior point method) to train HSTM, it can cause large time consumption due to its ample search space and iterative numerical calculations. In comparison, by replacing the original numerical optimization problem of large size with analytic sequential optimization problems of small size, the AHSTM oriented decomposition algorithm is presented to ensure the efficient training of the AHSTM.

Although the proposed AHSTM has achieved excellent performance, there are still two aspects to be studied in the future: 1) embed the kernel technology into AHSTM to further improve the classification performance and 2) extend the AHSTM to address multiple classification problems directly.

REFERENCES

[1] F. T. Mahmoudi, F. Samadzadegan, and P. Reinartz, "Object recognition based on the context aware decision-level fusion in multiviews imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 1, pp. 12–22, Jan. 2015.

[2] B. Wu, X. Sun, Q. Wu, M. Yan, H. Wang, and K. Fu, "Building reconstruction from high-resolution multiview aerial imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 4, pp. 855–859, Apr. 2015.

[3] H. Zhang, H. Lin, and Y. Li, "Impacts of feature normalization on optical and SAR data fusion for land use/land cover classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 5, pp. 1061–1065, May 2015.

[4] M. Zhou, Z. Zou, Z. Shi, W.-J. Zeng, and J. Gui, "Local attention networks for occluded airplane detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 3, pp. 381–385, Mar. 2020.

[5] J. Pang, C. Li, J. Shi, Z. Xu, and H. Feng, "R^2-CNN: Fast tiny object detection in large-scale remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5512–5524, Aug. 2019.

[6] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[7] J. Jiao *et al.*, "A densely connected end-to-end neural network for multiscale and multiscene SAR ship detection," *IEEE Access*, vol. 6, pp. 20881–20892, 2018.

[8] G. Sumbul, R. G. Cinbis, and S. Aksoy, "Multisource region attention network for fine-grained object recognition in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4929–4937, Jul. 2019.

[9] V. Vapnik, "Pattern recognition using generalized portrait method," *Automat. Remote Control*, vol. 24, no. 6, pp. 774–780, Jan. 1963.

[10] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, 1995.

[11] D. Li, S. Du, and T. Wu, "A weighted support vector machine method and its application," in *Proc. 5th World Congr. Intell. Control Autom.*, Jun. 2004, pp. 1834–1837.

[12] J. A. K. Suykens, T. V. J. Gestel, D. B. Brabanter, D. Moor, and J. Vandewalle, *Least Squares Support Vector Machines*. Singapore: World Scientific, 2002.

[13] B. Schölkopf, A. J. Smola, R. C. Williamson, and P. L. Bartlett, "New support vector algorithms," *Neural Comput.*, vol. 12, no. 5, pp. 1207–1245, May 2000.

[14] G. Lantzanakis, Z. Mitraka, and N. Chrysoulakis, "X-SVM: An extension of C-SVM algorithm for classification of high-resolution satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 3805–3815, May 2021.

[15] D. Tao, X. Li, W. Hu, S. Maybank, and X. Wu, "Supervised tensor learning," in *Proc. 5th IEEE Int. Conf. Data Mining (ICDM)*, Nov. 2005, pp. 450–457.

[16] X. Geng, K. Smith-Miles, Z.-H. Zhou, and L. Wang, "Face image modeling by multilinear subspace analysis with missing values," *IEEE Trans. Syst., Man, Cybern. B, Cybern.*, vol. 41, no. 3, pp. 881–892, Jun. 2011.

[17] H. Lu, K. N. Plataniotis, and A. N. Venetsanopoulos, "Uncorrelated multilinear discriminant analysis with regularization and aggregation for tensor object recognition," *IEEE Trans. Neural Netw.*, vol. 20, no. 1, pp. 103–123, Jan. 2009.

[18] D. Tao, X. Li, X. Wu, W. Hu, and S. J. Maybank, "Supervised tensor learning," *Knowl. Inf. Syst.*, vol. 13, no. 1, pp. 1–42, 2007.

[19] Z. Wang and S. Chen, "New least squares support vector machines based on matrix patterns," *Neural Process. Lett.*, vol. 26, no. 1, pp. 41–56, Jul. 2007.

[20] Z. Hao, L. He, B. Chen, and X. Yang, "A linear support higher-order tensor machine for classification," *IEEE Trans. Image Process.*, vol. 22, no. 7, pp. 2911–2920, Jul. 2013.

[21] Z.-Y. Chen, Z.-P. Fan, and M. Sun, "A multi-kernel support tensor machine for classification with multitype multiway data and an application to cross-selling recommendations," *Eur. J. Oper. Res.*, vol. 255, no. 1, pp. 110–120, Nov. 2016.

[22] B. Zhou, B. Song, M. M. Hassan, and A. Alamri, "Multilinear rank support tensor machine for crowd density estimation," *Eng. Appl. Artif. Intell.*, vol. 72, pp. 382–392, Jun. 2018.

[23] X. Guo, X. Huang, L. Zhang, L. Zhang, A. Plaza, and J. A. Benediktsson, "Support tensor machines for classification of hyperspectral remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3248–3264, Jun. 2016.

[24] H. Lu, K. N. Plataniotis, and A. N. Venetsanopoulos, "MPCA: Multilinear principal component analysis of tensor objects," *IEEE Trans. Neural Netw.*, vol. 19, no. 1, pp. 18–39, Jan. 2008.

[25] Z. Ma, L. T. Yang, and Q. Zhang, "Support multimode tensor machine for multiple classification on industrial big data," *IEEE Trans. Ind. Informat.*, vol. 17, no. 5, pp. 3382–3390, May 2021.

[26] X. Liu, T. Guo, L. He, and X. Yang, "A low-rank approximation-based transductive support tensor machine for semisupervised classification," *IEEE Trans. Image Process.*, vol. 24, no. 6, pp. 1825–1838, Jun. 2015.

[27] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev.*, vol. 51, no. 3, pp. 455–500, 2009.

[28] I. Kotsia, W. Guo, and I. Patras, "Higher rank support tensor machines for visual recognition," *Pattern Recognit.*, vol. 45, pp. 4192–4203, Dec. 2012.

[29] I. Kotsia and I. Patras, "Support Tucker machines," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2011, pp. 633–640.

[30] C.-C. Chang and C.-J. Lin, "Training v-support vector classifiers: Theory and algorithms," *Neural Comput.*, vol. 13, no. 9, pp. 2119–2147, Sep. 2001.

[31] P. Savicky and J. Vomlel, "Exploiting tensor rank-one decomposition in probabilistic inference," *Kybernetika*, vol. 43, no. 5, pp. 747–764, 2007.

[32] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge Univ. Press, 2004.

[33] J. C. Platt, "Fast training of support vector machines using sequential minimal optimization," in *Advances in Kernel Methods—Support Vector Learning*, B. Schölkopf, C. J. C. Burges, and A. J. Smola, Eds. Cambridge, MA: MIT Press, 1998.

[34] J. A. Momoh, S. X. Guo, E. C. Ogbuobiri, and R. Adapa, "The quadratic interior point method solving power system optimization problems," *IEEE Trans. Power Syst.*, vol. 9, no. 3, pp. 1327–1336, Aug. 1994.

[35] R. A. Bartlett and L. T. Biegler, "QPSchur: A dual, active-set, Schur-complement method for large-scale and structured convex quadratic programming," *Optim. Eng.*, vol. 7, no. 1, pp. 5–32, Mar. 2006.

[36] Q. Wu *et al.*, "Online transfer learning with multiple homogeneous or heterogeneous sources," *IEEE Trans. Knowl. Data Eng.*, vol. 29, no. 7, pp. 1494–1507, Jul. 2017, doi: 10.1109/TKDE.2017.2685597.

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[37] X. Li, L. Zhang, B. Du, L. Zhang, and Q. Shi, "Iterative reweighting heterogeneous transfer learning framework for supervised remote sensing image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 2022–2035, May 2017.
[38] Y. Luo, Y. Wen, and D. Tao, "Heterogeneous multitask metric learning across multiple domains," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 9, pp. 4051–4064, Sep. 2018.
[39] L. P. Zhang, L. Zhang, D. Tao, and X. Huang, "A multifeature tensor for remote-sensing target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 2, pp. 374–378, Feb. 2011.
[40] T. Ojala, M. Pietikäinen, and D. Harwood, "A comparative study of texture measures with classification based on featured distributions," *Pattern Recognit.*, vol. 29, no. 1, pp. 51–59, Jan. 1996.
[41] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2005, vol. 1, no. 1, pp. 886–893, doi: 10.1109/CVPR.2005.177.
[42] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 3rd ed. Upper Saddle River, NJ, USA: Prentice-Hall, 2006.
[43] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.
[44] H. Su, S. Maji, E. Kalogerakis, and E. Learned-Miller, "Multi-view convolutional neural networks for 3D shape recognition," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 945–953, doi: 10.1109/ICCV.2015.114.
[45] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.



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